

Depression and anxiety in adults with epilepsy. Systematic literature review and meta-analysis

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ABSTRACT

Background: People with epilepsy (PWE) are at increased risk of psychiatric comorbidities, particularly depression and anxiety. These conditions adversely affect adherence, quality of life, and seizure control, yet reported frequencies and correlates vary widely. We aimed to estimate the occurrence of depressive and anxiety symptoms in PWE and summarize associated demographic and clinical variables.

Methods: We searched five databases for studies published between 2018 and 2024. Observational studies assessing symptoms or diagnoses of depression and/or anxiety in adults with epilepsy using validated instruments were included. Study selection, data extraction, and quality appraisal (JBI) were performed independently by multiple reviewers. Random-effects meta-analyses (REML) were conducted using cross-sectional, screening-based proportions.

Results: Fifty-four studies were included for data extraction; 49 reported depression outcomes and 33 anxiety outcomes (some reported both). Meta-analysis included 12 cross-sectional studies of depressive symptoms ($n = 3866$; 1358 screen-positive) and 14 cross-sectional studies of anxiety symptoms ($n = 4321$; 1515 screen-positive). The pooled proportion was 40% for depression (95% CI: 30–50%; $I^2 = 97.54\%$; $Q = 427.70$, $p < 0.001$) and 36% for anxiety (95% CI: 28–44%; $I^2 = 96.64\%$; $Q = 250.60$, $p < 0.001$). Both outcomes were frequently associated with female sex, polytherapy, temporal lobe epilepsy, and higher seizure frequency; drug-resistant epilepsy and poor seizure control were also commonly reported.

Conclusion: Depressive and anxiety symptoms are frequent in PWE, supporting systematic screening and integrated mental health care within epilepsy management.

1. Introduction

Epilepsy is defined according to the International League Against Epilepsy [ILAE 2014] as a disorder of the brain characterized by an enduring predisposition to generate epileptic seizures (Fisher et al., 2014). These seizures arise from an excessively synchronous and sustained discharge of a group of neurons caused by a persistent state of excitability (Sánchez Fernández et al., 2019).

A 2024 systematic analysis of the Global Burden of neurologic diseases found that the global prevalence of epilepsy is 308.9 per 100,000 (Steinmetz et al., 2024). However, the prevalence of epilepsy in low-

income countries is still underestimated, with reports suggesting figures close to 7 per 1000 people (Hajji et al., 2024).

Given its complex pathophysiology, lifelong or chronic condition and frequent polypharmacy, epilepsy is usually a condition of remarkable comorbidities (Manole et al., 2023). For over 150 years, it has been reported that mental health conditions, particularly depression and anxiety disorders, are more common in people with epilepsy than in the general population. In 1880, William Gowers (1845–1915) recognized interictal cognitive and behavioral disorders, which he believed were a result of the underlying seizures and medications (Eadie, 2011).

According to the Diagnostic and Statistical Manual of Mental

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Disorders in its fifth version (DSM-5), depression may be experienced as anhedonia, disturbance of sleep or appetite, changes in psychomotor activity, trouble with tasks that require concentration or making decisions, feelings of guilt or depressed mood, fatigue and recurrent thoughts of death and suicide. Certain types of epilepsy also involve common networks responsible for mood expression and regulation, especially when associated with frontal-temporal dysfunction, as seen in temporal lobe epilepsy (Berg et al., 2017).

Moreover, the progression of psychiatric comorbidities is also favored by the discrimination and stigmatization suffered by people with epilepsy (PWE) throughout history and even nowadays (Palacios Espinosa et al., 2024), which may predispose to social or family issues. In 1960 Samuel Livingston (–1984) stated in his book *Comprehensive Management of Epilepsy in Infancy, Childhood and Adolescence* “The most serious hazard of an epileptic disorder is frequently not the seizures, but the associated emotional disturbances which are prone to develop in a youngster as a result of mismanagement by his family, by his classmates and friends and by society” (Livingston, 1972). This may cause a negative reaction to the diagnosis and have an impact on their mental health, clinical course and prognosis.

Taking this into account, the screening of psychiatric comorbidities in epilepsy is thus important to give patients further care including additional pharmacological interventions or to make a referral path to mental health specialists promptly (Gandy et al., 2025). In this context, the present study aims to conduct a systematic review with meta-analysis to identify the occurrence of depression and anxiety disorders and symptoms in adults with a prior diagnosis of epilepsy, considering both diagnosed conditions and self-reported symptoms without formal diagnostic confirmation.

2. Methods

This systematic review was registered in the International Prospective Register of Systematic Reviews (PROSPERO ID: CRD42024608977). It was developed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Supplementary material 1).

2.1. Data search strategy

The following databases were consulted to identify eligible studies: PubMed, Scopus, Cochrane-Ovid, Academic Search Complete and Psychology and Behavioral Science. The search was conducted on November 11th, 2024, for all four databases. Full search strategies for the databases are presented below (Table 1). Duplicates were identified and removed using Rayyan.

2.2. Eligibility criteria

Studies were considered eligible if they were published in peer-reviewed scientific journals from 2018 onward, due to last systematic review of the topic including articles until 2018, included adult patients (≥18 years) with a confirmed clinical diagnosis of epilepsy, reported outcomes of depression or anxiety, also were included if reported clinical diagnosis of depression or anxiety using any assessment instrument described in the study, adopted observational designs (cohort or cross-sectional studies), and were published in English or Spanish.

Studies were excluded if they focused on pediatric or adolescent populations (under 18 years old), or included mixed populations without separate data for adults, showed poor methodological quality that precluded reliable data analysis to identify the frequency of report of depression and anxiety disorders and symptoms in adults with a prior diagnosis of epilepsy, were published in languages other than English or Spanish or focused exclusively in veteran populations. Veterans with epilepsy represent a unique subgroup with distinct clinical, psychiatric, and social profiles, including a significantly higher burden of psychiatric

Table 1
Data search strategy per database.

Database	Search strategy	Number of results
PubMed	(anxiety OR anxiety disorder) AND (depression OR depressive disorder) AND (Epilepsy) AND (prevalence OR incidence) AND (adults OR adult) AND (comorbidities) Filter: 2018–2024	143
Scopus	(anxiety) AND (depression OR depressive disorder) AND (epilepsy) AND (prevalence) AND (adult) AND PUBYEAR >2017 AND PUBYEAR <2025 AND (LIMIT-TO (LANGUAGE, “English”) OR LIMIT-TO (LANGUAGE, “Spanish”)) AND (LIMIT-TO (EXACTKEYWORD, “Human”) OR LIMIT-TO (LANGUAGE, “Spanish”)) AND (LIMIT-TO (EXACTKEYWORD, “Human”) OR LIMIT-TO (EXACTKEYWORD, “Humans”))	192
Cochrane-OVID	(anxiety OR anxiety disorder) AND (depression OR depressive disorder) AND (Epilepsy) AND (prevalence OR incidence) AND (adults OR adult) AND (comorbidities)	12
Academic Search Complete	(anxiety OR anxiety disorder) AND (depression OR depressive disorder) AND (Epilepsy) AND (prevalence OR incidence) AND (adults OR adult) AND (comorbidities)	142
Psychology and Behavioral Science	(anxiety OR anxiety disorder) AND (depression OR depressive disorder) AND (Epilepsy) AND (prevalence OR incidence) AND (adults OR adult) AND (comorbidities)	12

comorbidities such as PTSD, depression, and anxiety, largely influenced by military-related trauma and a higher prevalence of traumatic brain injury (Faught et al., 2024; Quiroga-Padilla et al., 2022), these factors introduce a set of exposures and vulnerabilities not generalizable to the broader population with epilepsy.

2.3. Selection process

All retrieved articles were uploaded to Rayyan for duplicate removal and screening. Study selection was performed by all authors (JCRL, LPS, XPE, SNP, MAG, AMB). Title and abstract screening was conducted independently by paired reviewers, using Rayyan's blinding option so that reviewers could not see each other's decisions. Disagreements were resolved by discussion and, when needed, adjudicated by a third reviewer from the author team. Following the initial screening, full-text review was conducted independently by a different pair of reviewers, with discrepancies resolved by a third team member. All studies meeting inclusion criteria underwent risk of bias assessment and were included for data extraction and qualitative synthesis (Fig. 1).

2.4. Assessment of methodological quality and risk of bias

Using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for 39 cross-sectional studies (Moola et al., 2020a). Of these, 13 studies (33%) received an overall judgment of moderate risk of bias, mainly due to incomplete reporting on confounder identification and strategies, while the remaining 26 (67%) were rated as low risk of bias. Overall adherence to JBI criteria was high, with a mean fulfillment of 91% (range 63–100%). For the remaining studies (12 cohort and 3 case-control), the specific JBI appraisal tools were applied (Moola et al., 2017; Moola et al., 2020b); these studies did not present critical methodological flaws and were consistently judged as low to moderate risk, without compromising the validity of the evidence. Further details of the individual assessments are available in Supplementary Material 2.

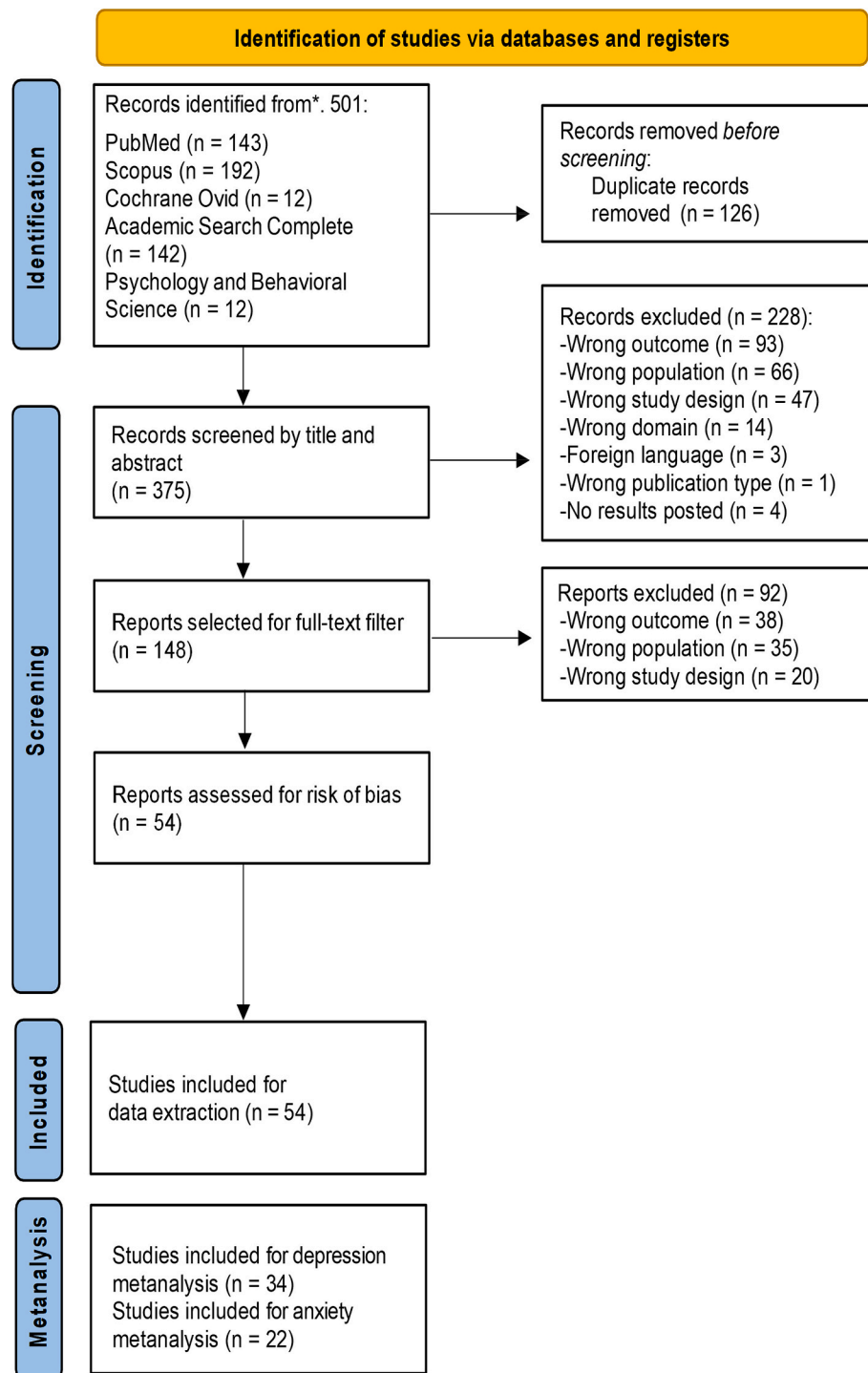


Fig. 1. PRISMA flow diagram.

2.5. Data collection process

Data collected from eligible studies was recorded in an Excel spreadsheet that included key demographic and clinical information. Specifically, we extracted the following variables: frequency of anxiety or depression (whether reported as a primary or secondary outcome). We also documented the assessment instruments described in each study for measuring anxiety and depression, associated factors, the presence of any control group comparisons (if available), and the general results of each study. Data extraction was performed by one reviewer and independently corroborated by another, without the use of automation tools.

2.6. Narrative and Statistical analysis of depression and anxiety

We describe the general characteristics of the included studies narratively, and we summarize the main findings by grouping studies by condition (depression vs. anxiety). Throughout the Results and Discussion, we use the terms “depressive symptoms” and “anxiety symptoms” to refer to screening-based estimates, and “clinical depression” or “clinical anxiety” to refer to diagnosis-based estimates. In the sections below, the terms “depression” and “anxiety” are used in a broader sense to include both screening-based symptoms and clinical diagnoses, unless otherwise specified; the assessment approach is stated whenever

relevant to ensure clarity.

We conducted two separate meta-analyses: one for depressive symptoms and one for anxiety symptoms. Both were exploratory in nature, aimed at generating initial quantitative estimates of the proportion of each condition in people with epilepsy. Given the marked variability in how studies reported their data.

and the relatively small number that met consistent methodological standards, an exploratory approach was considered appropriate.

Studies were included in the MA if they met three criteria: (Fisher et al., 2014) they used a cross-sectional design (Sánchez Fernández et al., 2019) they applied a screening instrument to assess depressive symptoms or anxiety symptoms and (Steinmetz et al., 2024) they reported enough information to calculate a proportion, specifically the total sample size and the number of individuals who screened positive. All eligible studies reported only overall proportions and did not stratify results by epilepsy subtype. Consequently, these samples were analyzed together under the category “unspecified epilepsy”. This category reflects the original reporting of the primary studies, including both unstratified data (where authors did not report results by epilepsy subtype) and population that may have inherently included heterogeneous or unclassifiable epilepsy syndromes.

Pooled proportions with 95% confidence intervals (CIs) were estimated using random-effects models. Heterogeneity was quantified using the I^2 statistic, and all analyses were performed with Stata 18 (StataCorp, 2023).

3. Results

3.1. Study selection

501 studies were retrieved (143 from PubMed, 192 from Scopus, 12 from Cochrane Ovid, 142 from Academic search complete and 12 from Psychology and Behavioral Science) published between 2018 and 2024. Then, 218 duplicates were removed. After screening titles and abstracts 228 articles were excluded, leaving 147 articles for full-text review. Of these, 92 were excluded due to inappropriate population, outcome, or study design, resulting in 54 articles included for data extraction. Overall, 49 studies contributed data on depression and 33 studies contributed data on anxiety, with some articles reporting both outcomes. For the extraction tables, some publications appear in more than one row because they reported results for distinct subgroups or populations within the same article; however, for the descriptive characteristics and study counts, each publication was considered only once (Fig. 1). All studies were published in English. Among the included studies, 12 cross-sectional studies on depressive symptoms and 14 cross-sectional studies on anxiety symptoms met eligibility criteria and were included in the meta-analysis, while the remaining studies were summarized narratively.

3.2. General characteristics of included studies

We present below the characteristics of the included studies, categorized separately for depression and anxiety. Summary tables are provided in Table 2 (Depression) and Table 3 (Anxiety). In these tables, the data reflect the total study sample size and the overall number of identified cases, without stratification by epilepsy subtype. Additional details are provided in the Supplementary Material, including subtype-stratified information, instrument characteristics, how each instrument was used in the study (screening vs diagnostic purpose), and whether diagnostic confirmation was performed after a positive screening result (Supplementary Material 3 and 4).

3.2.1. Depression

For depression ($n = 49$), cross-sectional studies were the predominant design ($n = 38$), followed by cohort studies ($n = 4$) and case-control studies ($n = 4$). One study used a prospective longitudinal design ($n = 1$)

and one used a before-and-after design ($n = 1$); in one study, the design was not clearly reported ($n = 1$). Across the 49 depression studies (counting each publication once, despite duplicated extraction rows for subgroup reporting), a total of 14,679 participants were included. Sex-disaggregated data were reported in 44 studies ($n = 13,042$), with near-equal representation of males ($n = 6398$) and females ($n = 6642$); five studies did not provide usable sex totals in the extracted table (four NR and one with inconsistent counts).

Studies were conducted primarily in Asia ($n = 23$; 47%), followed by America ($n = 9$; 18%), Africa ($n = 8$; 16%), Europe ($n = 8$; 16%), and Oceania ($n = 1$; 2%). The largest contributions came from China ($n = 8$) and the United States ($n = 7$), followed by India ($n = 4$) and Turkey ($n = 3$). Ethiopia, Iran, Spain, France, and Nigeria each contributed two studies ($n = 2$ each). Single studies were conducted in Poland, Morocco, Pakistan, South Korea, Thailand, Germany, Canada, the United Kingdom, Nepal, Australia, Rwanda, Egypt, Guinea, Malaysia, Indonesia, and Brazil. In addition, one multi-country European study included Belgium, Spain, Italy, France, the UK, and Germany (Table 2).

We classified countries according to World Bank income categories. Studies were conducted most frequently in high-income (HIC; $n = 18$; 37%) and upper-middle-income settings (UMIC; $n = 17$; 35%), followed by lower-middle-income (LMIC; $n = 9$; 18%) and low-income countries (LIC; $n = 5$; 10%).

Most studies recruited institution-based samples ($n = 43$; 88%), with epilepsy-focused institution-based populations being the most frequent source ($n = 25$; 51%). Institution-based neurology populations accounted for 10 studies (20%), and the remaining institution-based samples included more general clinical settings ($n = 7$; 14%) and one neuropsychiatric population ($n = 1$; 2%). Population-based studies were less common ($n = 5$; 10%), and one study used a national cohort-based design ($n = 1$; 2%).

Most studies assessed depressive symptoms using screening instruments ($n = 31$), while 14 studies relied exclusively on diagnostic ascertainment. Four studies reported both screening-based and diagnostic outcomes within the same publication (e.g. Gandhi et al., 2022; Scott et al., 2021; Talibov et al., 2024; Tandiawan et al., 2023). Overall, instruments were predominantly general mental health measures ($n = 41$; 84%), whereas a smaller subset used epilepsy-specific tools ($n = 8$; 16%), almost exclusively in a screening format.

The most frequently used depression measures were NDDI-E/C-NDDI-E ($n = 10$) and PHQ-9 ($n = 9$), followed by HADS-D ($n = 4$) and BDI/BDI-II ($n = 4$). Instrument language/version was reported in 30 studies, most commonly Chinese ($n = 7$) and English ($n = 5$), with additional validated versions in Persian and Turkish ($n = 2$ each) and several other local languages. Diagnostic ascertainment most commonly relied on the MINI ($n = 8$) and SCID-based interviews ($n = 3$). A smaller number of studies used other approaches, including HAM-D/HDRS ($n = 4$), DASS ($n = 2$), SCL-90R ($n = 2$), SDS ($n = 1$), PHQ-2 ($n = 1$), and CIDI ($n = 1$), and a minority relied on ICD-10-coded diagnoses ($n = 2$). Several publications applied more than one instrument within the same study; therefore, these instrument counts are not mutually exclusive. When screening instruments were applied, cut-offs varied across studies (e.g., PHQ-9 ≥ 5 to ≥ 10 and > 6 ; HADS-D > 7 to ≥ 8 ; NDDI-E ≥ 11 to > 15 ; BDI thresholds from ≥ 14 to ≥ 20 ; HAM-D > 6 to ≥ 8 ; SDS ≥ 53 ; PHQ-2 ≥ 3), while cut-offs were not applicable for interview- or record-based diagnoses. Administration was most often self-administered ($n = 16$), followed by trained staff/interviewer administration ($n = 10$), psychiatrist-led assessment ($n = 8$), mixed administration formats ($n = 5$), and clinician-led assessment ($n = 3$); administration details were not reported in six studies ($n = 6$), and one study relied on medical-record coding ($n = 1$).

3.2.2. Anxiety

For anxiety ($n = 33$), cross-sectional studies were the predominant design ($n = 23$), followed by cohort studies ($n = 9$). One study reported anxiety comorbidity data in the context of a randomized controlled trial

Table 2

Characteristics of studies reporting depression in people with epilepsy (n = 49).

Depression cases are reported as screening-positive and/or diagnosis-confirmed. Some publications appear in multiple rows because they reported separate depression estimates for the same underlying sample using different instruments, thresholds, or ascertainment approaches (e.g., screening vs diagnostic); however, each article was counted once in the study totals. Epilepsy type abbreviations: F = focal; G = generalized; M = mixed; F-BG = focal-to-bilateral tonic-clonic; U = unspecified. (NR = not reported).

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Abe.2020	USA	HIC	BDI-II English and Spanish versions	Cohort study	56	57	S	G	BDI-II ≥ 20 (moderate to severe)	Self-administered	No	–	113	BDI-II 12 $\pm$ 10	TLE BDI-II 12 $\pm$ 10
Abe.2020	USA	HIC	BDI-II English and Spanish versions	Cohort study	56	57	S	G	BDI-II ≥ 20 (moderate to severe)	Self-administered	No	–	113	BDI-II 12 $\pm$ 10	ETLE BDI-II 11 $\pm$ 7
Abe.2020	USA	HIC	BDI-II English and Spanish versions	Cohort study	56	57	S	G	BDI-II ≥ 20 (moderate to severe)	Self-administered	No	–	113	BDI-II 12 $\pm$ 10	Nonfocal NR
Abe.2020	USA	HIC	BDI-II English and Spanish versions	Cohort study	56	57	S	G	BDI-II ≥ 20 (moderate to severe)	Self-administered	No	–	113	PHQ-9: 8 $\pm$ 6	PHQ-9: TLE 8 + –6
Abe.2020	USA	HIC	BDI-II English and Spanish versions	Cohort study	56	57	S	G	BDI-II ≥ 20 (moderate to severe)	Self-administered	No	–	113	PHQ-9: 8 $\pm$ 6	PHQ-9: ETLE 7 + –6
Addis.2021	Ethiopia	LIC	HADS Amharic version	Cross-sectional study	205	165	S	G	HADS-D ≥ 8	Interview-administered (staff)	No	Perceived stigma (AOR 3.89; 95% CI 2.27–6.68). Low educational status (AOR 0.48; 95% CI 0.25–0.92). Rural residence (AOR 0.50; 95% CI 0.28–0.89). High seizure frequency (AOR 2.07; 95% CI 1.01–4.23). Poor social support (AOR 2.73; 95% CI 1.41–5.31)	370	137	NR
Ajinkya.2020	United States	HIC	PHQ-2 Language not reported	Cross-sectional study	849	1175	S	G	PHQ-2 ≥ 3	Self-administered questionnaire (MEPS)	No	Female sex (OR 1.40; 95% CI 1.05–1.87). Age 35–49 vs 18–34 (OR 1.83; 95% CI 1.23–2.72). Age 50–64 vs 18–34 (OR 1.47; 95% CI 0.95–2.28). Age ≥ 65 vs 18–34 (OR 0.82; 95% CI 0.53–1.27). Charlson Comorbidity Index ≥ 1 (OR	2024	517	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Munger Clary.2018	USA	HIC	SCL-90R English or Spanish version	Cross-sectional study	242	297	S	G	SCL-90R depression T-score ≥ 60	Self-administered	No	1.92; 95% CI 1.41–2.61). Higher family income ($\geq 400\%$ FPL) (OR 0.33; 95% CI 0.21–0.51) NR	540	297	NR
Asadi-Pooya.2024	Iran	UMIC	DASS-21 Persian version	Cross-sectional study	22	58	S	G	DASS-21 depression severity categories (cut-off for extracted count NR)	Self-administered	No	–	80	22	Focal 9/40
Asadi-Pooya.2024	Iran	UMIC	DASS-21 Persian version	Cross-sectional study	22	58	S	G	DASS-21 depression severity categories (cut-off for extracted count NR)	Self-administered	No	–	80	22	IGE 13/40
Benoist.2023	Belgium, Spain, Italy, France, UK, Germany	HIC	ICD-10 D codes	Cohort study	NR	NR	D	G	Not applicable (ICD-10 diagnosis)	Clinician coding in electronic medical records	No	–	1075	287	Focal 287/1075
Campos-Fernández, 2020	Spain	HIC	DSM-5 (mood disorder)	Cross-sectional study	77	80	D	G	Not applicable (DSM-5)	Clinician during outpatient visit	No	Seizure frequency. $r = -0.193$, $p = 0.053$. Irritability. $r = 0.216$, $p = 0.039$. Depression severity. $r = -0.598$, $p < 0.001$	157	HADS-D 6.9 ± 4.5	NR
Dabla.2020	India	LMIC	HADS Language not reported	Cross-sectional study	70	30	S	G	HADS-D > 7	Self-administered	No	Female gender ($p = 0.004$). Polytherapy ($p = 0.0002$). Uncontrolled epilepsy ($p < 0.001$). Drug resistant epilepsy ($p < 0.001$). Temporal lobe epilepsy ($p < 0.004$)	100	60	TLE 14/15
Dabla.2020	India	LMIC	HADS Language not reported	Cross-sectional study	70	30	S	G	HADS-D > 7	Self-administered	No	Female gender ($p = 0.004$). Polytherapy ($p = 0.0002$). Uncontrolled	100	60	Non-TLE 46/85

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Forthoffer.2021	France	HIC	NDDI-E Language not reported	Cross-sectional study	50	62	S	Sp	NDDI-E ≥ 15	Self-administered	No	epilepsy ($p < 0.001$). Drug resistant epilepsy ($p < 0.001$). Temporal lobe epilepsy ($p < 0.004$)	112	14	Focal 11/87
Forthoffer.2021	France	HIC	NDDI-E Language not reported	Cross-sectional study	50	62	S	Sp	NDDI-E ≥ 15	Self-administered	No	–	112	14	Generalized 3/22
Gandhi et al., 2022	India	LMIC	MINI Language not + DASS	Case-control study	77	63	D	G	Not applicable	NR	No	–	140	MINI 43	NR
Gandhi et al., 2022	India	LMIC	MINI Language not + DASS	Case-control study	77	63	S	G	Not applicable	NR	No	–	140	DASS 47	NR
Goel.2022	India	LMIC	NDDI-E Punjabi translated version	Cross-sectional study	NR	NR	S	Sp	NDDI-E > 15	Study nurse supervised self-administered at home	No	Education below high school (OR 2.59; 95% CI 1.12–5.10; $p = 0.03$). Later age of seizure onset (OR 1.05; 95% CI 1.00–1.10; $p = 0.04$). Seizure ≥ 1 /year (OR 2.36; 95% CI 1.00–5.58; $p = 0.05$). Clobazam use (OR 5.09; 95% CI 1.40–18.42; $p = 0.01$)	116	17	NR
Grzegorzewska.2024	Poland	HIC	SCID-I (DSM-IV-TR)	Cross-sectional study	31	65	D	G	Not applicable	Single researcher (MSW)	No	–	96	NR	NR
Hajji.2024	Morocco	LMIC	HADS Language not reported	Cross-sectional study	140	154	S	G	HADS-D ≥ 8	Investigator-administered face-to-face interview	No	Antiseizure polytherapy (aOR 3.35; 95% CI 1.72–6.54). Anxiety (aOR 24.04; 95% CI 12.12–47.67)	294	132	NR
Ho.2020	China	UMIC	ICD-10 D criteria		291	267	D	G	Not applicable (ICD-10 diagnosis)	Psychiatrist	No	Female sex. OR 3.56 (95% CI 1.36–9.36), $p = 0.01$. Drug-resistant epilepsy. OR 4.79 (95% CI 2.06–11.12), $p < 0.001$.	558	44	Focal aware 1/16; Focal impaired awareness 40/454; Generalized onset 3/81; Unspecified 0/7

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Jansen.2019	France	HIC	NDDI-E Language not reported	Cross-sectional study	39	48	S	Sp	NR	2 psychiatrists	No	Clonazepam use. OR 14.41 (95% CI 4.20–49.47), $p < 0.001$. Valproate use. OR 0.37 (95% CI 0.15–0.91), $p = 0.03$	87	15	Right 8/40; Left 9/38; Bilateral/ nonlateralized NR
Kanner.2023	USA	HIC	MINI Language not reported	Cross-sectional study	142	205	D	G	Not applicable	Trained research assistant	No	–	347	33	Focal 33/347
Katyal.2021	India	LMIC	MINI English or Hindi versions	Cohort study	162	155	D	G	Not applicable	NR	No	Seizure during pandemic, OR 2.6 (95% CI 1.45–4.73). Increased seizure frequency, OR 1.9 (95% CI 1.01–3.57). Altered use of ASMs, OR 2.9 (95% CI 1.19–7.24). Previous history of depression, OR 8.8 (95% CI 4.54–17.21)	317	63	Focal 23/111; Generalized 40/206
Keskin.2023	Turkey	UMIC	BDI	Cross-sectional study	25	58	S	G	BDI ≥ 17 defined but n not reported	Self-administered questionnaires under clinician supervision	No	–	44	BDI 14.02 $\pm$ 9.66	NR
Keskin.2023	Turkey	UMIC	BDI	Cross-sectional study	25	58	S	G	BDI ≥ 17 defined but n not reported	Self-administered questionnaires under clinician supervision	No	–	44	BDI 14.02 $\pm$ 9.66	NR
Khalid.2024	Pakistan	LMIC	HAM-D Language not reported	Cross-sectional study	NR	NR	S	G	NR	Certified psychiatrist	No	Antiseizure polytherapy vs monotherapy. OR 1.352 (95% CI 1.129–1.619), $p = 0.010$	120	31.70%	NR
Kim.2019	South Korea	HIC	PHQ-9 Korean context	Cross-sectional study	29	37	S	G	PHQ-9 ≥ 10	Self-administered (self-reported questionnaires)	No	–	66	17	NR
Kuladee.2019	Thailand	UMIC	MINI Thai version	Cross-sectional study	70	100	D	G	Not applicable	NR (face-to-face interview)	No	Temporal lobe EEG abnormalities. AOR 4.01 (95% CI	170	17	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Kumar.2019	USA	HIC	PHQ-9 English version	Cross-sectional study	38	81	S	G	PHQ-9 ≥ 10	Self-administered in-person at study medical center	Yes	1.47–10.92), $p = 0.007$ Inability to work ($p = 0.05$). Seizures in past 30 days ($p < 0.01$). Seizure count in past 30 days ($p = 0.002$). Lower quality of life ($p < 0.0001$). Bipolar disorder ($p = 0.02$). Panic disorder ($p < 0.01$). OCD ($p < 0.01$)	120	62	NR
Labudda.2018	Germany	HIC	MINI (DSM-IV)	Case-control study	63	57	D	G	Not applicable	Trained interviewers (authors and additional interviewers)	No	–	120	26	NR
Li.2021	China	UMIC	PHQ-9 (Chinese version)	Before-after design	164	130	S	G	PHQ-9 > 6	Self-administered (online questionnaire via Sojump/WeChat)	No	Lockdown: changes in drug regimen (OR 2.738; 95% CI 1.209–6.198); uncontrolled seizures (OR 2.182; 95% CI 1.180–4.034). Reopening: seizure frequency ≥ 2 /month (OR 2.153; 95% CI 1.102–4.206); worry about in-person medical consultation (OR 1.924; 95% CI 1.073–3.450); low life satisfaction (OR 4.021; 95% CI 2.325–6.955). Increased PHQ-9 score: high education level (OR 1.946; 95% CI 1.191–3.182); low life satisfaction (OR 1.940; 95% CI 1.007–3.737).	294	Lockdown COVID 22.5%;	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Li.2021	China	UMIC	PHQ-9 (Chinese version)	Before-after design	164	130	S	G	PHQ-9 > 6	Self-administered (online questionnaire via Sojump/WeChat)	No	(Two-wave study: depression 22.5% lockdown = 66/294; 34.0% reopening = 100/294.) Lockdown: changes in drug regimen (OR 2.738; 95% CI 1.209–6.198); uncontrolled seizures (OR 2.182; 95% CI 1.180–4.034). Reopening: seizure frequency ≥ 2/month (OR 2.153; 95% CI 1.102–4.206); worry about in-person medical consultation (OR 1.924; 95% CI 1.073–3.450); low life satisfaction (OR 4.021; 95% CI 2.325–6.955). Increased PHQ-9 score: high education level (OR 1.946; 95% CI 1.191–3.182); low life satisfaction (OR 1.940; 95% CI 1.007–3.737). (Two-wave study: depression 22.5% lockdown = 66/294; 34.0% reopening = 100/294.) Males. Fewer education years (OR 0.769). Lower social function (OR 0.698). Low energy/fatigue (OR 0.482). Females. Low	294	Reopening Post COVID34.0%	Focal
Liu.2020	China	UMIC	SDS Chinese version	Cross-sectional study	93	65	S	G	SDS ≥53	Resident doctors assisted participants with self-report questionnaires	Yes		158	58	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Madandola.2020	Nigeria	LMIC	CIDI Language not reported	Cross-sectional study	266	134	D	G	Not applicable	One researcher trained by a CIDI certified trainer	No	energy/fatigue (OR 0.265) Minority ethnic group ($p = 0.006$). Family history of mental illness ($p = 0.021$). GHQ caseness ($p \leq 0.001$)	400	35	NR
Mc Carthy.2024	USA	HIC	PHQ-9 (English)	Cross-sectional study	21	34	S	G	PHQ-9 ≥ 10	Trained staff (clinical research coordinator; participant completed on iPad/REDCap)	Yes	No statistically significant association between PHQ-9 category (≥ 10 vs < 10) and social support impact, financial impact, or employment impact (Fisher's exact tests). (Alternate count also reported: PHQ-9 $\geq 5 = 29/55$.)	55	17	NR
Monteagudo-Gimeno.2023	Spain	HIC	SCID interview Language not reported	Cross-sectional study	NR	NR	D	G	Not applicable	Psychiatrist	No	Interictal dysphoric disorder. OR 8.40 (95% CI 4.34–16.88), $p < 0.001$	282	64	IDD 46/98; Seizure type: Focal 69.5%; Generalized 24.8%; No data 5.7%. Locus: Temporal 48.2%; Extratemporal 22.0%; Unestablished 18.1%. Etiology: Idiopathic 63.8%; Secondary 36.2%.
Monteagudo-Gimeno.2023	Spain	HIC	SCID interview Language not reported	Cross-sectional study	NR	NR	D	G	Not applicable	Psychiatrist	No	Interictal dysphoric disorder. OR 8.40 (95% CI 4.34–16.88), $p < 0.001$	282	64	No IDD 18/184 Seizure type: Focal 69.5%; Generalized 24.8%; No data 5.7%. Locus: Temporal 48.2%; Extratemporal 22.0%; Unestablished

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Munger Clary. 2023* (Owolabi et al., 2016)	USA	HIC	SCL-90R Language not reported	Cross-sectional study	181	239	S	G	NR	Self-report (completed during routine epilepsy neuropsychology evaluation)	No	NR	420	T score 60.2 (11.4)	18.1%. Etiology: Idiopathic 63.8%; Secondary 36.2%. NR
Ogundare.2020	Nigeria	LMIC	MINI Plus (Depression module; DSM-IV/ICD-10)	Cross-sectional study	147	123	D	G	Not applicable	Interviewer-administered	No	Younger age ($\chi^2 = 11.808$; $df = 2$; $p = 0.002$). Single marital status ($\chi^2 = 9.787$; $df = 3$; $p = 0.016$). Income < N5,000 ($\chi^2 = 12.077$; $df = 4$; $p = 0.012$). Younger age at onset ($\chi^2 = 3.890$; $df = 1$; $p = 0.049$). Very frequent seizures in past month (OR 5.104; $p = 0.014$; also $\chi^2 = 19.668$; $df = 4$; $p = 0.001$). Frequent seizures in past month (OR 3.266; $p = 0.029$). Uncontrolled seizures ($\chi^2 = 5.83$; $df = 1$; $p = 0.012$).	270	32	NR
Prisnie.2018	Canada	HIC	PHQ-9 English version	Cross-sectional study	126	153	S	G	NR	Self-administered questionnaire	No	–	279	(mean, SD) 6.3 [5.8]	NR
Roberts-West.2023	UK	HIC	HADS English context	Cross-sectional study	93	118	S	G	HADS-D ≥ 8 (n derived from reported percentages)	NR	No	–	211	Focal 33%, Depression moderate: 12%, Severe: 4%, mild: 14%	NR
Roberts-West.2023	UK	HIC	HADS English context	Cross-sectional study	93	118	S	G	HADS-D ≥ 8 (n derived from reported percentages)	NR	No	–	211	Generalized 28%, Depression mild: 7%, moderate18%	NR
Roberts-West.2023	UK	HIC	HADS English context	Cross-sectional study	93	118	S	G	HADS-D ≥ 8 (n derived from	NR	No	–	211	Unclassified 29% Depression mild: 17%,	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Saeedi.2024	Iran	UMIC	BDI Persian version	Case-control study	35	41	S	G	reported percentages) BDI ≥ 14 (mild or worse)	Self-administered questionnaire (researchers assisted if illiterate)	No	Longer disease duration. OR 1.128 (95% CI 1.110–1.729), $p = 0.002$	76	moderate: 12% 54	NR
Sah.2020	Nepal	LIC	HAM-D Nepali version	Cross-sectional study	87	55	S	G	HAM-D ≥ 8	Self-administered questionnaire	No	Polytherapy vs monotherapy. OR 3.82 (95% CI 1.61–9.05), $p = 0.002$	142	44	GTCS 35/105
Sah.2020	Nepal	LIC	HAM-D Nepali version	Cross-sectional study	87	55	S	G	HAM-D ≥ 8	Self-administered questionnaire	No	Polytherapy vs monotherapy. OR 3.82 (95% CI 1.61–9.05), $p = 0.002$	142	44	PS 9/37
Sair.2021	Turkey	UMIC	SCID-I (DSM-IV Axis I; Turkish version)	Case-control study	8	22	D	G	Not applicable	Psychiatry (clinician-administered by psychiatrist)	No	Depression not related to disease duration ($p = 0.796$) or age at TLE diagnosis ($p = 0.241$); anxious temperament higher in those with depression ($p = 0.038$).	30	22	Temporal lobe epilepsy 22/30
Scott.2021	Australia	HIC	MINI (DSM-5) + NDDI-E (English)	Cross-sectional study	113	266	D	G	NDDI-E > 15 ; Not applicable (MINI diagnosis)	Mixed (clinician interview; self-administered online)	No	–	379	Hospital (MINI-defined depressive disorder) $n = 30/106$;	NR
Scott.2021	Australia	HIC	MINI (DSM-5) + NDDI-E (English)	Cross-sectional study	113	266	S	Sp	NDDI-E > 15 ; Not applicable (MINI diagnosis)	Mixed (clinician interview; self-administered online)	Yes	–	379	Community (NDDI-E > 15) $n = 143/273$	NR
Sebera. 2018 (Monteagudo-Gimeno et al., 2023)	Rwanda	LIC	PHQ-9 (Kinyarwanda version); HDRS/ Hamilton Depression Rating Scale (Kinyarwanda version)	Prospective longitudinal study	335	237	S	G	PHQ-9 > 5 ; HDRS ≥ 6 (any depression)	Trained staff (interviewer-administered)	No	–	572	Any depression (HDRS) 81/572; MSD (HDRS > 17) 27/572; MSD in 12-month period 46/572	MSD (HDRS > 17): Focal 23/248
Sebera. 2018 (Monteagudo-Gimeno et al., 2023)	Rwanda	LIC	PHQ-9 (Kinyarwanda version); HDRS/ Hamilton	Prospective longitudinal study	335	237	S	G	PHQ-9 > 5 ; HDRS ≥ 6 (any depression)	Trained staff (interviewer-administered)	No	–	572	Any depression (HDRS) 81/572; MSD (HDRS > 17)	MSD (HDRS > 17): Generalized 10/100

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Sebera. 2018 (Monteagudo-Gimeno et al., 2023)	Rwanda	LIC	Depression Rating Scale (Kinyarwanda version) PHQ-9 (Kinyarwanda version); HDRS/ Hamilton Depression Rating Scale (Kinyarwanda version)	Prospective longitudinal study	335	237	S	G	PHQ-9 > 5; HDRS ≥6 (any depression)	Trained staff (interviewer-administered)	No	–	572	27/572; MSD in 12-month period 46/572 Any depression (HDRS) 81/ 572; MSD (HDRS >17) 27/572; MSD in 12-month period 46/572	MSD (HDRS >17);Unknown 13/211
Sebera. 2018 (Monteagudo-Gimeno et al., 2023)	Rwanda	LIC	PHQ-9 (Kinyarwanda version); HDRS/ Hamilton Depression Rating Scale (Kinyarwanda version)	Prospective longitudinal study	335	237	S	G	PHQ-9 > 5; HDRS ≥6 (any depression)	Trained staff (interviewer-administered)	No	–	572	Any depression (HDRS) 81/ 572; MSD (HDRS >17) 27/572; MSD in 12-month period 46/572	MSD (HDRS >17); Unclassified 0/ 13
Sehlo.2024	Egypt	LMIC	PHQ-9 (validated Arabic version)	Cross-sectional study	152	138	S	G	PHQ-9 ≥ 5	Clinician (doctor interview)	Yes	Not working — OR 5.83; Low financial status — OR 11.90; Refractory epilepsy — OR 14.60; Fear of COVID-19 infection — OR 5.06; Fear of death (COVID-19) — OR 4.99; Death of close person — OR 3.65; Sleep disturbance — OR 12.71; Decreased family support — OR 3.95; Increased seizure rate — OR 7.10; Increased ER visits — OR 5.76; Lack of drug adherence — OR 4.27; Lack of routine follow-up — OR 3.75	290	204	NR
Seid.	Ethiopia	LIC	PHQ-9 (questionnaire)	Cross-sectional study	163	133	S	G	PHQ-9 > 9	Trained staff (interviewer-administered)	No	Illiteracy (unable to read/write) — AOR 0.40 (95% CI	296	103	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
			translated into Tigrigna)									0.16–0.99); Long ASM use ≥ 11 years — AOR 3.72 (95% CI 1.50–9.21); No improvement with ASMs — AOR 0.22 (95% CI 0.10–0.46); Perceived stigma — AOR 0.24 (95% CI 0.13–0.46); Stress symptoms — AOR 0.45 (95% CI 0.22–0.93)			
Shi.2023	China	UMIC	C-NDDI-E (Chinese version of NDDI-E)	Cross-sectional study	691	635	S	Sp	C-NDDI-E > 12	Self-administered (QR-code smartphone questionnaire)	Yes	Focal or focal-to-bilateral TCSZ — OR 1.531 (95% CI 1.094–2.138), $p = 0.013$. Seizure occurrence in the last 3 months — OR 1.644 (95% CI 1.130–2.411), $p = 0.010$. Age (per 1-year increment) — OR 0.962 (95% CI 0.944–0.980), $p = 4.12e-5$.	1326	362	NR
Song.2024	China	UMIC	NDDI-E (Chinese version)	Cross-sectional study	208	188	S	Sp	NDDI-E > 13	Self-administered (self-report questionnaire)	Yes	NR	396	114	NR
Sylla.2020	Guinea	LIC	PHQ-9 (language of participant's preference; administered in French and/or local language)	Cross-sectional study	76	64	S	G	PHQ-9 ≥ 5	Clinician (Guinean resident physicians; in-person interview)	Yes	Seizure in the past month — OR 3.03 (95% CI 2.63–3.48), $p = 0.01$; Gender, stigma, injuries, and number of ASMs — not significant ($p > 0.05$)	140	93	NR
Talıbov.2024	Turkey	UMIC	MINI (Turkish; DSM-IV/ICD-10) Turk	Cross-sectional study	126	174	D	G	Not applicable	Psychiatry (psychiatrists)	No	High seizure frequency — $p = 0.01$; Number of ASMs — $p = 0.007$; Seizure in previous 12 months — $p = 0.003$; History of epilepsy surgery	300	47 (MINI major depression, diagnosis-confirmed);	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Tahbov.2024	Turkey	UMIC	MINI (Turkish; DSM-IV/ICD-10) Turk	Cross-sectional study	126	174	S	G	Not applicable	Psychiatry (psychiatrists)	Yes	— $p = 0.032$; History of psychiatric disease — OR 3.133, $p < 0.001$; Higher QOLIE-10 score — OR 2.634, $p < 0.001$; Higher stigma scale score — OR 1.485, $p < 0.001$ High seizure frequency — $p = 0.01$; Number of ASMs — $p = 0.007$; Seizure in previous 12 months — $p = 0.003$; History of epilepsy surgery — $p = 0.032$; History of psychiatric disease — OR 3.133, $p < 0.001$; Higher QOLIE-10 score — OR 2.634, $p < 0.001$; Higher stigma scale score — OR 1.485, $p < 0.001$	300	122 (SCL-90-R GSI ≥ 1 , screen-positive)	NR
Tan.2021	Malaysia	UMIC	DSM-5 questionnaire	Cross-sectional study	67	62	D	D	Not applicable	Psychiatry (psychiatrists)	No	Low socioeconomic background ($p = 0.026$); generalized epilepsy ($p = 0.036$); temporal lobe epilepsy ($p = 0.010$); unemployed/non-white-collar jobs ($p = 0.046$)	129	12 (DSM-5 MDD, diagnosis-confirmed)	Generalized motor 7/57;
Tan.2021	Malaysia	UMIC	DSM-5 questionnaire	Cross-sectional study	67	62	D	D	Not applicable	Psychiatry (psychiatrists)	No	Low socioeconomic background ($p = 0.026$); generalized epilepsy ($p = 0.036$); temporal lobe epilepsy ($p =$	129	12 (DSM-5 MDD, diagnosis-confirmed)	Generalized non-motor 4/21;

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Tan.2021	Malaysia	UMIC	DSM-5 questionnaire	Cross-sectional study	67	62	D	D	Not applicable	Psychiatry (psychiatrists)	No	0.010); unemployed/non-white-collar jobs (p = 0.046) Low socioeconomic background (p = 0.026); generalized epilepsy (p = 0.036); temporal lobe epilepsy (p = 0.010); unemployed/non-white-collar jobs (p = 0.046)	129	12 (DSM-5 MDD, diagnosis-confirmed)	Focal motor 1/46;
Tan.2021	Malaysia	UMIC	DSM-5 questionnaire	Cross-sectional study	67	62	D	D	Not applicable	Psychiatry (psychiatrists)	No	Low socioeconomic background (p = 0.026); generalized epilepsy (p = 0.036); temporal lobe epilepsy (p = 0.010); unemployed/non-white-collar jobs (p = 0.046)	129	12 (DSM-5 MDD, diagnosis-confirmed)	Focal to bilateral tonic-clonic 0/2;
Tan.2021	Malaysia	UMIC	DSM-5 questionnaire	Cross-sectional study	67	62	D	D	Not applicable	Psychiatry (psychiatrists)	No	Low socioeconomic background (p = 0.026); generalized epilepsy (p = 0.036); temporal lobe epilepsy (p = 0.010); unemployed/non-white-collar jobs (p = 0.046)	129	12 (DSM-5 MDD, diagnosis-confirmed)	Unknown 0/3
Tandiawan.2023	Indonesia	UMIC	NDDI-E (Indonesian version) + DSM-5 questionnaire	Cross-sectional study	45	58	S	G	NDDI-E ≥ 11	NR	Yes	–	103	Depressive symptoms (NDDI-E ≥ 11) 19/103;	Depressive symptoms (NDDI-E ≥ 11) 19/103;
Tandiawan.2023	Indonesia	UMIC	NDDI-E (Indonesian version) + DSM-5 questionnaire	Cross-sectional study	45	58	D	G	MDD (DSM-5)	NR	No	–	103	MDD (DSM-5) 7/103	TLE 7/81

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
Tandiawan.2023	Indonesia	UMIC	NDDI-E (Indonesian version) + DSM-5 questionnaire	Cross-sectional study	45	58	D	G	MDD (DSM-5)	NR	No	–	103	MDD (DSM-5) 0/12	IGE 0/12
Tandiawan.2023	Indonesia	UMIC	NDDI-E (Indonesian version) + DSM-5 questionnaire	Cross-sectional study	45	58	D	G	MDD (DSM-5)	NR	No	–	103	MDD (DSM-5) 0/10	Unknown 0/10
Zhong.2021	China	UMIC	C-NDDI-E (Chinese version of NDDI-E)	Cohort study	340	234	S	Sp	C-NDDI-E > 12	Self-administered (self-rated questionnaire); Psychiatry (psychiatrists)	No	Males: Seizure frequency over the past year — OR 1.586 (95% CI 1.019–2.468), <i>p</i> = 0.041; Females: No positive associations reported (OR > 1 not observed)	157	C-NDDI-E > 12: Baseline 39/157C-NDDI-E > 12 after12 months 35/157	12 months: Focal 27/119
Zhong.2021	China	UMIC	C-NDDI-E (Chinese version of NDDI-E)	Cohort study	340	234	S	Sp	C-NDDI-E > 12	Self-administered (self-rated questionnaire); Psychiatry (psychiatrists)	No	Males: Seizure frequency over the past year — OR 1.586 (95% CI 1.019–2.468), <i>p</i> = 0.041; Females: No positive associations reported (OR > 1 not observed)	157	C-NDDI-E > 12: Baseline 39/157C-NDDI-E > 12 after12 months 35/157	12 months: Generalized 5/24
Zhong.2021	China	UMIC	C-NDDI-E (Chinese version of NDDI-E)	Cohort study	340	234	S	Sp	C-NDDI-E > 12	Self-administered (self-rated questionnaire); Psychiatry (psychiatrists)	No	Males: Seizure frequency over the past year — OR 1.586 (95% CI 1.019–2.468), <i>p</i> = 0.041; Females: No positive associations reported (OR > 1 not observed)	157	C-NDDI-E > 12: Baseline 39/157C-NDDI-E > 12 after12 months 35/157	12 months: Unclassified 3/14
Zhong.2021	China	UMIC	C-NDDI-E (Chinese version of NDDI-E)	Cross-sectional study	91	66	S	Sp	C-NDDI-E > 12	Self-administered	Yes	MoCA <26 at baseline — OR 2.419 (95% CI 1.093–5.350), <i>p</i> = 0.029; MoCA <26 at 12 months — OR 3.007 (95% CI 1.223–7.390), <i>p</i> = 0.016; Depressive symptoms at baseline →	574	116	NR

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Table 2 (continued)

Author- Year	Population origin	World Bank 2024	Instrument language/ version	Study design	Gender male (n)	Gender female (n)	Scale use in the study (screening or Diagnostic)	General (G) or epilepsy-specific instrument (Sp)	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Depression cases by epilepsy subtype
												depressive symptoms at 12 months — OR 4.539 (95% CI 1.973–10.445), p < 0.001			
Zhu.2022	China	UMIC	HAM-D (Chinese version)	Cross-sectional study	73	58	S	G	HAM-D > 6	NR	Yes	–	131	48	NR
Zingano.2019	Brazil	UMIC	DSM-IV-TR psychiatric interview	Cross-sectional study	44	59	D	G	Not applicable	Psychiatry (psychiatrist)	No	–	103	24	MTLE-HS 24/ 103

Table 3
 Characteristics of studies reporting anxiety in people with epilepsy (n = 33).
 Anxiety cases are reported as screening-positive and/or diagnosis-confirmed. Some publications appear in multiple rows because they reported separate anxiety estimates for the same underlying sample using different instruments, thresholds, or ascertainment approaches (e.g., screening vs diagnostic); however, each article was counted once in the study totals. Epilepsy type abbreviations: F = focal; G = generalized; M = mixed; F-BG = focal-to-bilateral tonic-clonic; U = unspecified. (NR = not reported).

Author- year	Population origin	World Bank 2024	Anxiety scale language or version	Study design	Gender male (n)	Gender female (n)	Scale purpose (screening or diagnostic)	General or epilepsy-specific instrument	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Anxiety cases by epilepsy subtype
Abe,2020	USA	HIC	GAD-7 (English & Spanish versions)	Cross-sectional study	61	52	Screening	General	≥10	Self-administered	No	–	113	GAD-7 9 ± 7	TLE GAD-7 9 ± 6
Abe,2020	USA	HIC	GAD-7 (English & Spanish versions)	Cross-sectional study	61	52	Screening	General	≥10	Self-administered	No	–	113	GAD-7 9 ± 7	ETLE GAD-7 10 ± 8
Abe,2020	USA	HIC	GAD-7 (English & Spanish versions)	Cross-sectional study	61	52	Screening	General	≥10	Self-administered	No	–	113	GAD-7 9 ± 7	Nonfocal NR
Adjei,2021	Ghana	LMIC	BSI (53-item)	Cross-sectional study	78	88	Screening	General	NR	Self-administered	No	–	166	BSI Anxiety: Males 1.58 (0.67); Females 1.95 (0.95)	BSI Anxiety: Males 1.58 (0.67); Females 1.95 (0.95)
Asadi-Pooya,2024	Iran	UMIC	DASS-21 (validated Farsi version)	Cross-sectional study	22	58	Screening	General	NR	Self-administered	No	–	80	Any anxiety 51/80; Normal 29/80; Mild 11/80; Moderate 15/80; Severe 5/80; Extremely severe 20/80	Focal epilepsy: Any anxiety 27/40; Normal 13/40; Mild 6/40; Moderate 8/40; Severe 2/40; Extremely severe 11/40
Asadi-Pooya,2024	Iran	UMIC	DASS-21 (validated Farsi version)	Cross-sectional study	22	58	Screening	General	NR	Self-administered	No	–	80	Any anxiety 51/80; Normal 29/80; Mild 11/80; Moderate 15/80; Severe 5/80; Extremely severe 20/80	IGE: Any anxiety 24/40; Normal 16/40; Mild 5/40; Moderate 7/40; Severe 3/40; Extremely severe 9/40
Benoist,2023	Belgium/ Spain/ Italy/ France/ UK/ Germany	HIC	ICD-10 (EMR-coded anxiety diagnosis)	Cohort	NR	NR	Diagnostic	General	NA	Clinician (medical record coding)	No	–	1075	127	Focal drug-resistant epilepsy 127/ 1075 (11.8%)
Campos-Fernández,2020	Spain	HIC	DSM-5 clinical evaluation; HADS	Cross-sectional study	77	80	Screening	General	NR	Clinician	Yes	–	157	Anxiety (medical records) n = 25; HADS–Anxiety 8.6 ± 4.9	25
Campos-Fernández,2020	Spain	HIC	DSM-5 clinical evaluation; HADS	Cross-sectional study	77	80	Screening	General	NR	Clinician	Yes	–	157	Anxiety (medical records) n = 25; HADS–Anxiety 8.6 ± 4.9	HADS–Anxiety 8.6 ± 4.9
Dabla,2020	India	LMIC	HADS-A (NR language/version)	Cross-sectional study	70	30	Screening	General	>7	Self-administered	Yes	Polytherapy — p = 0.004; Uncontrolled	100	70	Idiopathic n = 46/73

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Table 3 (continued)

Author- year	Population origin	World Bank 2024	Anxiety scale language or version	Study design	Gender male (n)	Gender female (n)	Scale purpose (screening or diagnostic)	General or epilepsy-specific instrument	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Anxiety cases by epilepsy subtype
Dabla,2020	India	LMIC	HADS-A (NR language/version)	Cross-sectional study	70	30	Screening	General	>7	Self-administered	Yes	epilepsy — $p < 0.001$; Drug-resistant epilepsy — $p < 0.0004$; Temporal lobe epilepsy (TLE) — $p < 0.004$ Polytherapy — $p = 0.004$; Uncontrolled epilepsy — $p < 0.001$; Drug-resistant epilepsy — $p < 0.0004$; Temporal lobe epilepsy (TLE) — $p < 0.004$	100	70	Symptomatic $n = 14/27$
Forthoffer,2021	France	HIC	GAD-7 (French)	Cohort	50	62	Screening	General	>7	NR	No	–	112	39	Focal $n = 28/87$
Forthoffer,2021	France	HIC	GAD-7 (French)	Cohort	50	62	Screening	General	>7	NR	No	–	112	39	Generalized $n = 8/22$
Gandhi,2022	India	LMIC	DASS-42 (Anxiety subscale; NR language); MINI (DSM-5/ICD-10)	Cross-sectional study	77	63	Diagnostic	General	NR; Not applicable	NR	No	–	140	36	DASS-42 $n = 36$
Gandhi,2022	India	LMIC	DASS-42 (Anxiety subscale; NR language); MINI (DSM-5/ICD-10)	Cross-sectional study	77	63	Diagnostic	General	NR; Not applicable	NR	No	–	140	36	MINI $n = 36$
Goel, 2022	India	LMIC	GAD-7 (vernacular version; NR) + ICD-10 (psychiatrist diagnosis)	Cross-sectional study	NR	NR	Screening	General	>6; Not applicable	Mixed (study nurse; neuropsychologist; psychiatrist)	No	–	116	Screening (GAD-7 > 6): 32	NR
Goel, 2022	India	LMIC	GAD-7 (vernacular version; NR) + ICD-10 (psychiatrist diagnosis)	Cross-sectional study	NR	NR	Screening	General	>6; Not applicable	Mixed (study nurse; neuropsychologist; psychiatrist)	Yes	–	116	Diagnostic (ICD-10 F40–F48): 17	NR
Hajji,2024	Morocco	LMIC	HADS (Moroccan dialect version)	Cross-sectional study	140	154	Screening	General	≥ 8	Interviewer-administered (investigator; face-to-face)	No	Depression — aOR 24.20 (95% CI 12.45–47.01)	294	151	Focal $n = 16/26$
Hajji,2024	Morocco	LMIC	HADS (Moroccan dialect version)	Cross-sectional study	140	154	Screening	General	≥ 8	Interviewer-administered (investigator; face-to-face)	No	Depression — aOR 24.20 (95% CI 12.45–47.01)	294	151	Generalized $n = 65/148$
Jansen,2019	France	HIC	MINI (DSM-IV; NR version) + GAD-7 (NR version)	Cohort	39	48	Diagnostic	General	NR; Not applicable	Mixed (psychiatrists interview; self-questionnaire)	No	Temporal lobe epilepsy vs	87	25 (MINI current anxiety disorder,	NR

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Table 3 (continued)

Author- year	Population origin	World Bank 2024	Anxiety scale language or version	Study design	Gender male (n)	Gender female (n)	Scale purpose (screening or diagnostic)	General or epilepsy-specific instrument	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Anxiety cases by epilepsy subtype
Jansen,2019	France	HIC	MINI (DSM-IV; NR version) + GAD-7 (NR version)	Cohort	39	48	Screening	General	NR; Not applicable	Mixed (psychiatrists interview; self-questionnaire)	No	extratemporal — p = 0.012 Temporal lobe epilepsy vs extratemporal — p = 0.012	87	diagnosis-confirmed) 33 (GAD-7 positive score, screen-positive)	NR
Kanner,2023	USA	HIC	MINI International Neuropsychiatric Interview (short version 5.0; DSM-IV-TR)	Cohort	142	205	Diagnostic	General	NA	Trained staff (research assistant)	No	—	347	95	NR
Khalid,2024	Pakistan	LMIC	HAM-A (NR version)	Cross-sectional study	NR	NR	Screening	General	NR	NR	Yes	—	120	19	NR
Kim,2019	Korea	HIC	GAD-7 (Korean validated version)	Cohort	29	37	Screening	General	≥10	Self-administered	No	—	66	15	NR
Kuladee,2019	Thailand	UMIC	MINI Thai version 5.0.0 (DSM-IV/ ICD-10)	Cross-sectional study	70	100	Diagnostic	General	NA	NR	No	—	170	9	NR
Kumar,2019	USA	HIC	Self-reported mental health comorbidity checklist (anxiety)	Description of comorbidity in a Randomized controlled trial in PWE	38	81	Screening	General	NA	Self-administered (in-person at study center)	No	—	120	39	NR
Li, 2021	China	UMIC	GAD-7 (NR language)	Cohort	164	130	Screening	General	>6	Self-administered (online questionnaire)	No	Lockdown period Unexpected seizure worries — OR 3.319 (95% CI 1.608–6.849) Medication regimen changes — OR 2.887 (95% CI 1.139–7.315) Low life satisfaction — OR 3.008 (95% CI 1.476–6.130) Reopening period Uncontrolled seizures — OR 2.251 (95% CI 1.056–4.800) Seizure frequency ≥ 2/ month — OR	294	53	NR

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Table 3 (continued)

Author- year	Population origin	World Bank 2024	Anxiety scale language or version	Study design	Gender male (n)	Gender female (n)	Scale purpose (screening or diagnostic)	General or epilepsy-specific instrument	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Anxiety cases by epilepsy subtype
Liu,2020	China	UMIC	SAS (Chinese version)	Cross-sectional study	93	65	Screening	General	≥50	Trained staff (resident doctors; interviewer-assisted)	Yes	2.153 (95% CI 1.102–4.206) Worry about in-person medical consultation — OR 1.924 (95% CI 1.073–3.450) High education level — OR 1.946 (95% CI 1.191–3.1 Males: Sleep quality OR 1.339, Energy/fatigue OR 0.432 Females: Seizure frequency OR 3.001, Energy/fatigue OR 0.151	158	50	NR
Mc Carthy,2024	USA	HIC	GAD-7 (English)	Cross-sectional study	21	34	Screening	General	≥8	Trained staff (clinical research coordinator; completed on iPad/ REDCap)	Yes	–	53	16	NR
Monteagudo-Gimeno,2023	Spain	HIC	SCID-IV (Spanish version) + HADS (Spanish version)	Cross-sectional study	NR	NR	Diagnostic	General	NA	Psychiatry (psychiatrist)	No	–	282	54	NR
Munger Clary,2018	USA	HIC	SCL-90R anxiety subscale (English/ Spanish)	Cross-sectional study	242	297	Screening	General	≥60	Self-administered	Yes	–	540	250	Generalized n = 17/59
Munger Clary,2018	USA	HIC	SCL-90R anxiety subscale (English/ Spanish)	Cross-sectional study	242	297	Screening	General	≥60	Self-administered	Yes	–	540	250	Focal n = 194/ 408
Munger Clary,2018	USA	HIC	SCL-90R anxiety subscale (English/ Spanish)	Cross-sectional study	242	297	Screening	General	≥60	Self-administered	Yes	–	540	250	Unknown n = 39/72
Munger Clary,2023	USA	HIC	SCL-90R phobic anxiety subscale (English/Spanish)	Cross-sectional study	180	239	Screening	General	≥60	Self-administered	No	–	420	153	Generalized n = 12/48
Munger Clary,2023	USA	HIC	SCL-90R phobic anxiety subscale (English/Spanish)	Cross-sectional study	180	239	Screening	General	≥60	Self-administered	No	–	420	153	Focal n = 121/ 317
Munger Clary,2023	USA	HIC	SCL-90R phobic anxiety subscale (English/Spanish)	Cross-sectional study	180	239	Screening	General	≥60	Self-administered	No	–	420	153	Mixed (both) NR
Munger Clary,2023	USA	HIC	SCL-90R phobic anxiety subscale (English/Spanish)	Cross-sectional study	180	239	Screening	General	≥60	Self-administered	No	–	420	153	Unknown n = 19/54
Prisnie,2018	Canada	HIC	HADS-A (English)	Cohort	126	153	Screening	General	NR	Self-administered (written questionnaire)	No	Higher HADS-A scores associated with lower SF-12	279	108	NR

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Table 3 (continued)

Author- year	Population origin	World Bank 2024	Anxiety scale language or version	Study design	Gender male (n)	Gender female (n)	Scale purpose (screening or diagnostic)	General or epilepsy-specific instrument	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Anxiety cases by epilepsy subtype
												Mental Component Score (MCS); Pratt index (variance contribution) 26.68%.			
Roberts-West, 2023	UK	HIC	HADS-A (English)	Cross-sectional study	93	118	Screening	General	≥8	Self-administered	Yes	–	211	126	Focal n = 100/165
Roberts-West, 2023	UK	HIC	HADS-A (English)	Cross-sectional study	93	118	Screening	General	≥8	Self-administered	Yes	–	211	126	Generalized n = 14/22
Roberts-West, 2023	UK	HIC	HADS-A (English)	Cross-sectional study	93	118	Screening	General	≥8	Self-administered	Yes	–	211	126	Unclassified n = 15/24
Rustamova, 2023	Uzbekistan	LMIC	HADS (NR language/version)	Cross-sectional study	33	27	Screening	General	NR	NR	Yes	–	60	28	NR
Scott,2021	Australia	HIC	brEASI (NR language/version) + MINI (DSM-5; NR version)	Cross-sectional study	113	266	Diagnostic	Diagnostic	>8; Not applicable	trained psychologist interview	No	–	379	33 (MINI anxiety disorder; diagnosis-confirmed)	NR
Scott,2021	Australia	HIC	brEASI (NR language/version) + MINI (DSM-5; NR version)	Cross-sectional study	113	266	Screening	Screening	>8; Not applicable	self-administered online	Yes	–	379	154 (brEASI >8; screen-positive)	NR
Shi,2023	China	UMIC	GAD-7 (Chinese version)	Cross-sectional study	691	635	Screening	General	>6	Self-administered (QR-code questionnaire)	Yes	Female sex — OR 1.467 (95% CI 1.134–1.899), p = 0.004. Focal or focal-to-bilateral TCSZ — OR 1.409 (95% CI 1.021–1.939), p = 0.036; Seizure in the last 3 months — OR 1.445 (95% CI 1.026–2.044), p = 0.036	1326	417	NR
Song,2024	China	UMIC	GAD-7 (Chinese version)	Cross-sectional study	208	188	Screening	General	>6	Self-administered	Yes	–	396	120	NR
Talibov,2024	Turkey	UMIC	SCL-90-R (Turkish) + MINI (Turkish; DSM-IV/ ICD-10)	Cohort	126	174	Screening	General	≥1; Not applicable	Mixed (self-report; psychiatry interview)	No	High seizure frequency p = 0.01; Number of ASMs p = 0.007; Seizure in previous 12	300	Screening: 122 (SCL-90-R GSI ≥ 1)	NR

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Table 3 (continued)

Author- year	Population origin	World Bank 2024	Anxiety scale language or version	Study design	Gender male (n)	Gender female (n)	Scale purpose (screening or diagnostic)	General or epilepsy-specific instrument	Cut-off (NR or value)	Who administered the instrument	Meta-analysis subset (Yes/ No)	Associated factors	Total Sample	Identified cases (n)	Anxiety cases by epilepsy subtype
Talibov,2024	Turkey	UMIC	SCL-90-R (Turkish) + MINI (Turkish; DSM-IV/ ICD-10)	Cohort	126	174	Diagnostic	General	≥1; Not applicable	Mixed (self-report; psychiatry interview)	No	months p = 0.003; History of epilepsy surgery p = 0.032; History of psychiatric disease OR 3.133 p < 0.001; High QOLIE-10 OR 2.634 p < 0.001; High stigma score OR 1.485 p < 0.001 High seizure frequency p = 0.01; Number of ASMs p = 0.007; Seizure in previous 12 months p = 0.003; History of epilepsy surgery p = 0.032; History of psychiatric disease OR 3.133 p < 0.001; High QOLIE-10 OR 2.634 p < 0.001; High stigma score OR 1.485 p < 0.001	300	Diagnostic: 16 (MINI GAD)	NR
Yetkin,2024	Turkey	UMIC	SCI-SAS + SASI + ASA-27 (NR language/version)	Cohort	41	64	Screening	General	Not applicable; NR	Psychiatry (psychiatrist)	No	–	105	31	NR
Zhong,2021	China	UMIC	GAD-7 (Chinese version)	Cross-sectional study	340	234	Screening	General	>6	Self-administered	Yes	–	574	162	Focal n = 139/ 480
Zhong,2021	China	UMIC	GAD-7 (Chinese version)	Cross-sectional study	340	234	Screening	General	>6	Self-administered	Yes	–	574	162	Not classified n = 23/94
Zhu,2022	China	UMIC	HAM-A (Chinese version)	Cross-sectional study	73	58	Screening	General	>6	NR	Yes	–	131	46	NR
Zingano,2019	Brazil	UMIC	DSM-IV-TR psychiatric interview + HADS-A + STAI-T (NR language/ version)	Cross-sectional study	44	59	Screening	General	>7; >53; Not applicable	Psychiatry (psychiatrist)	No	–	103	12	NR

in PWE ($n = 1$). Across the 33 anxiety studies (counting each publication once, despite duplicated extraction rows for subgroup reporting), a total of 8874 participants were included. Sex-disaggregated data were available for 25 studies ($n = 6148$), with near-equal representation of males ($n = 3000$) and females ($n = 3148$).

Studies were conducted primarily in Asia ($n = 16$; 48%), followed by America ($n = 8$; 24%), Europe ($n = 6$; 18%), Africa ($n = 2$; 6%), and Oceania ($n = 1$; 3%). The largest contributions came from the United States and China ($n = 6$ each), followed by India ($n = 3$), and Turkey, Spain, and France ($n = 2$ each). Single studies were conducted in Ghana, Iran, Pakistan, Korea, Thailand, Morocco, Canada, the United Kingdom, Uzbekistan, Australia, and Brazil. In addition, one multi-country European study included Belgium, Spain, Italy, France, the UK, and Germany (Table 3).

Countries were classified according to World Bank income categories. Anxiety studies were conducted most frequently in high-income country settings (HIC; $n = 15$; 45%), followed by upper-middle-income countries (UMIC; $n = 11$; 33%) and lower-middle-income countries (LMIC; $n = 7$; 21%). Most studies recruited institution-based samples ($n = 29$; 88%). Institution-based epilepsy populations were the most frequent source ($n = 21$; 64%), followed by institution-based neurology populations ($n = 8$; 24%). A smaller proportion of studies were population-based ($n = 4$; 12%).

Screening tools were used in most studies ($n = 28$), while diagnostic approaches were less frequent ($n = 8$). At the study level, 25 articles relied exclusively on screening instruments, 5 used diagnostic methods only, and 3 reported both screening and diagnostic assessments within the same publication (Scott et al., 2021; Talibov et al., 2024; Jansen et al., 2019). Overall, anxiety assessment relied almost entirely on general mental health instruments ($n = 32$; 97%), whereas only one study incorporated an epilepsy-specific tool ($n = 1$; 3%), used as a screening measure (brEASI) alongside diagnostic ascertainment.

Assessment was largely based on general anxiety instruments, with one study additionally using an epilepsy-focused tool (brEASI). The most frequently used scales were GAD-7 ($n = 10$) and HADS-A ($n = 4$), while diagnostic ascertainment commonly relied on MINI ($n = 4$). Instrument language/version was reported in 18 studies, most commonly Chinese ($n = 5$) and English ($n = 3$), with bilingual English/Spanish versions used in three studies and additional validated versions in French, Farsi, Turkish, Korean, Thai, Spanish, and a Moroccan dialect. A smaller number of studies used other approaches, including SCL-90-based measures ($n = 3$), SAS ($n = 1$), HAM-A ($n = 1$), and STAI-T ($n = 1$), and some studies combined screening tools with diagnostic criteria or coding (ICD-10 in $n = 1$; DSM-IV-TR in $n = 1$). When screening instruments were applied, cut-offs varied across studies (e.g., GAD-7 > 6 to ≥ 10 ; HADS-A > 7 to ≥ 8 ; brEASI > 8 ; SAS ≥ 50 ; HAM-A > 6 ; STAI-T > 53 ; SCL-90 thresholds), and cut-offs were not applicable in studies based on interviews or medical record diagnoses. Administration was most often self-administered ($n = 14$), followed by mixed administration formats ($n = 4$), psychiatrist-led assessment ($n = 3$), clinician/medical-record sources ($n = 2$), trained staff administration ($n = 2$), and interviewer-assisted administration ($n = 2$); administration details were not reported in six studies.

3.3. Frequency estimates and comparative results

3.3.1. Depression

Most cross-sectional studies reported depression as a proportion of PWE meeting either a screening threshold or a diagnosis. Among the 49 depression studies, 23 cross-sectional studies contributed prevalence estimates that could be mapped by epilepsy subtype and ascertainment method (Fig. 2). We present these proportions in a heatmap stratified by epilepsy type and by the type of ascertainment (diagnostic vs screening). For this visualization, epilepsy subtypes were grouped into focal, generalized, mixed, focal-to-bilateral tonic-clonic (F-BG), and an “unspecified” category. The latter included proportions reported without

further subtype specification, as well as studies that did not stratify depression outcomes by epilepsy type. Across studies assessing depression, epilepsy duration and participant age showed wide variability, with no consistent clustering around a single disease stage or age group (Table 2) (Supplementary Material 3).

Heatmap showing study-level proportions of depression in people with epilepsy, stratified by epilepsy type and displayed separately for diagnostic (Panel A) and screening instruments (Panel B). Epilepsy types are grouped as focal (F), generalized (G), mixed (M), focal-to-bilateral tonic-clonic (F-BG), and unspecified (U); values represent the percentage meeting the study-defined threshold or diagnosis.

In Panel A (diagnostic instruments), depression proportions ranged from 0% to 47%, with most values falling in the low-to-moderate range and predominantly reported under the “unspecified” category (approximately 9%–47%), while focal and generalized diagnostic estimates were generally lower (0%–23% and 0%–19%, respectively). In Panel B (screening instruments), proportions were wider and typically higher, ranging from 13% to 93% overall; focal screening estimates spanned 13%–93%, generalized estimates 14%–33%, and “unspecified” estimates 18%–70%. In studies contributing both diagnostic and screening estimates, screening-based proportions were consistently higher than diagnosis-based proportions. Across the heatmap, no consistent pattern suggested a clear predominance of depression in any single epilepsy subtype.

Within the case-control designs, two studies compared PWE with healthy adults and found a higher proportion of depression among cases (31% to 73%) than among controls (23% to 30%). One study used a Pure Psychogenic Nonepileptic Seizures (PNES) comparison group and reported a higher proportion of depression in controls than in cases (29% vs. 22%).

Regarding cohort and longitudinal studies, one study compared PWE with PNES patients and reported higher mean depression scores across epilepsy subtypes and instruments, including BDI-II in TLE and ETLE and PHQ-9. Another follow-up study reported depression proportions across epilepsy types, focal (22%), generalized (33%), and unclassified (36%), in a population with a 2-year epilepsy diagnosis, and at 12 months, there was a slight reduction in generalized and unclassified epilepsy. The European multicenter cohort reported a 27% occurrence of depression in focal epilepsy. One study assessed depression during lockdown and after post-COVID reopening, finding an increase from 22% to 34%, respectively. Finally, one study reported an outpatient follow-up with a baseline depression proportion of 14%, decreasing to 8% at 12 months (Supplementary Material 3).

3.3.2. Meta-analysis of depressive symptoms

The meta-analysis of depressive symptoms included 12 cross-sectional studies with a total of 3866 participants (Fig. 3). Individual study proportions ranged from 0.18 (95% CI: 0.11–0.26) in Tandiawan (2023) (Tandiawan et al., 2023) to 0.70 (95% CI: 0.65–0.76) in Sehlo (2024) (Sehlo et al., 2022). The pooled proportion under a random-effects model (REML) was 0.39 (95% CI: 0.30–0.47), with substantial heterogeneity across studies ($I^2 = 97.54\%$; $\tau^2 = 0.03$; $Q = 427.70$, $p < 0.001$). In the forest plot, each square represents a study-specific proportion, and its size reflects the study's weight, horizontal lines indicate 95% CIs, and the diamond represents the pooled estimate.

3.3.3. Anxiety

Most cross-sectional studies reported anxiety as a proportion of PWE meeting either a screening threshold or a diagnosis. Among the 33 anxiety studies, 22 cross-sectional studies contributed proportion estimates that could be mapped by epilepsy type and ascertainment method (Fig. 4). We present these proportions in a heatmap stratified by epilepsy type and by the type of ascertainment (diagnostic vs screening). For this visualization, epilepsy types were grouped into focal (F), generalized (G), mixed (M), and an “unspecified” category (U). The latter included proportions reported without further subtype specification, as well as

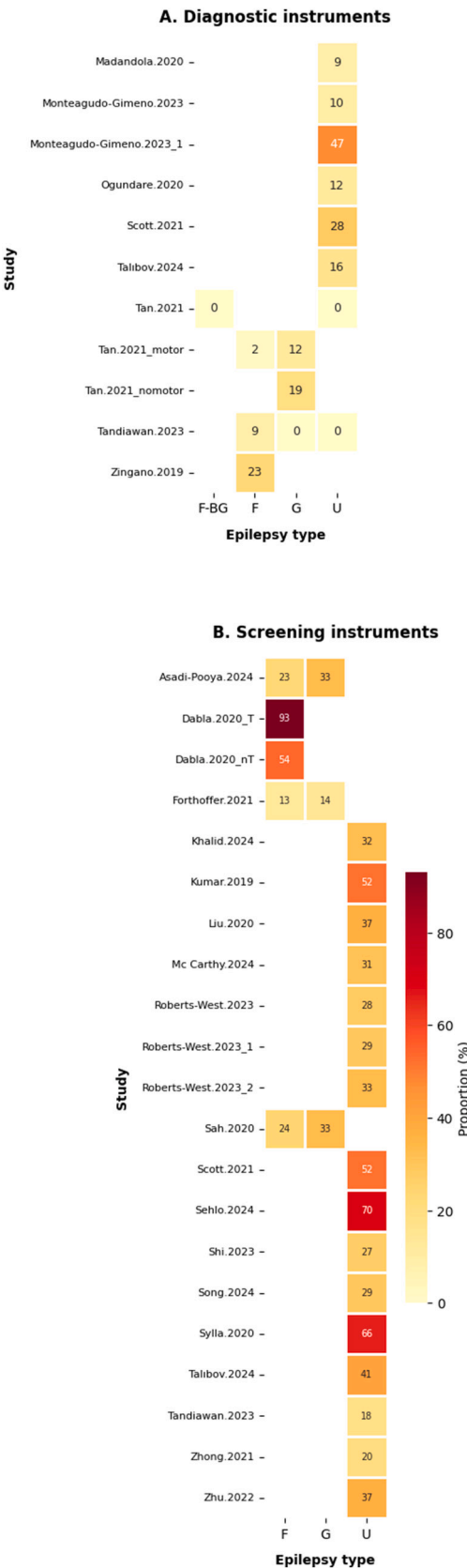


Fig. 2. Depression proportions by epilepsy type and ascertainment method (cross-sectional studies).

studies that did not stratify anxiety outcomes by epilepsy type. Consistent with this variability in reporting, epilepsy duration and participant age were also highly heterogeneous across anxiety studies, spanning a broad range of clinical trajectories and demographic profiles (Table 3) (Supplementary Material 4).

Heatmap showing study-level proportions of anxiety in people with epilepsy, stratified by epilepsy type and displayed separately for diagnostic (Panel A) and screening instruments (Panel B). Epilepsy types are grouped as focal (F), generalized (G), mixed (M), and unspecified (U); values represent the percentage meeting the study-defined screening threshold or diagnosis.

In Panel A (diagnostic instruments), anxiety proportions ranged from 5% to 26%, and all diagnostic estimates were reported under the “unspecified” category. In Panel B (screening instruments), proportions were wider and generally higher, ranging from 0% to 68% overall. Focal screening estimates ranged from 12% to 68%, generalized estimates from 25% to 64%, the mixed category showed a single 0% estimate, and “unspecified” screening estimates ranged from 15% to 63%. In the study contributing both approaches, screening yielded higher proportions than diagnosis-based ascertainment (Scott 2021: 41% vs 9%) (Scott et al., 2021).

A minority of studies reported continuous scale scores rather than proportions ($n = 3$). Abe et al. (2020) reported GAD-7 mean $\pm$ SD by focal subtype (TLE: 9 ± 6 ; ETLE: 10 ± 8 ; overall 9 ± 7) (Abe et al., 2020). Adjei et al. (2021) reported BSI Anxiety scores by sex (males: 1.58 ± 0.67 ; females: 1.95 ± 0.95) (Adjei et al., 2021). Campos-Fernández et al. (2020) reported HADS-Anxiety as a mean $\pm$ SD (8.6 ± 4.9) alongside anxiety documented in medical records ($n = 25$), without providing subtype-specific anxiety counts (Campos-Fernández et al., 2020).

Across the cohort studies, nine studies were identified, and four of them reported anxiety occurrence stratified by epilepsy type. In focal epilepsy, reported proportions ranged from 12% to 32%, while generalized epilepsy was reported at 36%. The remaining cohort studies were classified as unspecified epilepsy, with reported occurrence ranging from 5% to 38%.

3.3.4. Meta-analysis of anxiety symptoms

The meta-analysis for anxiety symptoms included 14 cross-sectional studies with a total of 4321 participants, of whom 1515 screened positive (Fig. 5). Individual study proportions ranged from 0.16 (Campos-Fernández et al., 2020; Khalid et al., 2024) to 0.70 (Dabla et al., 2020). The pooled proportion under a random-effects model (REML) was 0.36 (95% CI: 0.28–0.44). Considerable heterogeneity was observed across studies ($I^2 = 96.64\%$; $\tau^2 = 0.02$; $Q = 250.60$, $p < 0.001$).

3.4. Factors Associated with depression and anxiety

Across the included studies, higher seizure burden and lack of control were reported as associated with depression and/or anxiety in ten studies (Talibov et al., 2024; Goel et al., 2022; Katyal et al., 2021; Li et al., 2021; Liu et al., 2020; Ogundare et al., 2020; Sylla et al., 2020; Zhong et al., 2021). For example, very frequent seizures were associated with a fivefold increase in the odds of major depression ($OR = 5.10$, $p = 0.014$) (Ogundare et al., 2020), and pandemic-related seizure increases carried odds ratios between 2.1 and 7.1 (Sehlo et al., 2022; Katyal et al., 2021).

Female sex was found associated in three studies (Ajinkya et al., 2020; Ho et al., 2020; Shi et al., 2023). In one case-control study, women had more than threefold higher odds of depression ($OR = 3.56$; 95%CI: 1.36–9.36) (Ho et al., 2020), while a U.S. cohort reported a more modest association ($OR = 1.40$; 95%CI: 1.05–1.87) (Ajinkya et al., 2020).

Syndromic, structural, and electroclinical characteristics, including; mesial temporal sclerosis, temporal EEG abnormalities, structural epilepsy, and focal or focal-to-bilateral tonic-clonic seizures, were reported associated in five studies (Hajji et al., 2024; Ho et al., 2020; Shi et al.,

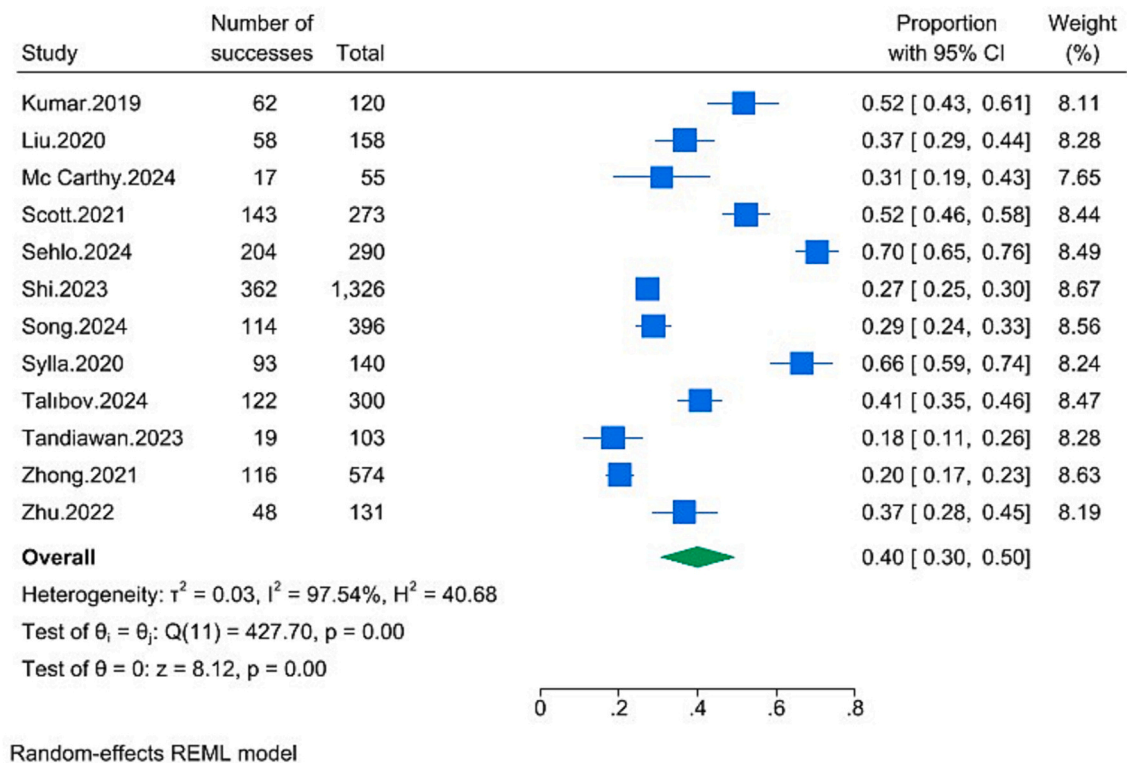


Fig. 3. Forest plot of the meta-analysis of depression symptoms proportions in individuals with epilepsy.

2023; Kuladee et al., 2019; Munger Clary et al., 2018). Mesial temporal sclerosis, for instance, was linked to higher anxiety (AOR = 2.12; 95%CI: 1.11–4.04) (Munger Clary et al., 2018), and temporal EEG abnormalities quadrupled the odds of psychiatric comorbidity (OR = 4.01; 95%CI: 1.47–10.92) (Kuladee et al., 2019). Drug-resistant epilepsy was described in one study with an almost fivefold increase (OR = 4.79; 95% CI: 2.06–11.12) (Ho et al., 2020).

Treatment-related findings included polytherapy and recent antiseizure-medication changes, reported associated in five studies (Hajji et al., 2024; Talibov et al., 2024; Khalid et al., 2024; Katyal et al., 2021; Li et al., 2021). In one large cross-sectional analysis, polytherapy was associated with a threefold increase in depression (AOR = 3.35; 95%CI: 1.72–6.54) (Hajji et al., 2024). Specific ASM exposures were highlighted: clonazepam use (OR = 14.41; 95%CI: 4.20–49.47) and clobazam (OR = 5.09; 95%CI: 1.40–18.42) (Goel et al., 2022; Ho et al., 2020). The history of epilepsy surgery was also associated (Talibov et al., 2024). Psychiatric or neurological comorbidity was reported associated in two studies (Talibov et al., 2024; Monteagudo-Gimeno et al., 2023). For example, interictal dysphoric disorder was linked to an almost eightfold higher prevalence of comorbid psychiatric diagnoses (OR = 7.88) (Monteagudo-Gimeno et al., 2023).

Sociodemographic and cognitive variables such as younger age or younger age at onset (Ogundare et al., 2020), longer disease duration (OR = 1.13 per year; 95%CI: 1.11–1.73) (Saeedi et al., 2024), lower educational attainment (OR = 2.59; 95%CI: 1.12–5.10) (Goel et al., 2022), cognitive deficits such as low MoCA scores (OR = 2.42–3.01) (Zhong et al., 2021) or lower IQ (OR = 1.36 per 10-point reduction; 95% CI: 1.18–1.57) (Munger Clary et al., 2023), non-White ethnicity (OR = 2.34; 95%CI: 1.12–4.84), and Spanish as native language (OR≈2.15–2.44) (Munger Clary et al., 2018; Munger Clary et al., 2023) were also reported associated.

Psychological and contextual factors included perceived stigma, which in one study nearly quadrupled the odds of depression (AOR = 3.89; 95%CI: 2.27–6.68) (83), and sleep disturbance, associated with higher risk in two studies (Sehlo et al., 2022; Liu et al., 2020). During

COVID-19, additional stressors and care disruptions, such as altered medication use (OR≈2.9), loss of follow-up (OR = 3.75), increased seizure activity (OR = 2.15–7.10), and fear of infection or death (OR≈5.0) were reported associated in three studies (Sehlo et al., 2022; Katyal et al., 2021; Li et al., 2021).

4. Discussion

This systematic review summarizes the frequency of comorbid depression and anxiety in adults with a prior diagnosis of epilepsy. A total of 54 studies were included in the data extraction process, with 49 reporting outcomes on depression and 33 on anxiety. Meta-analyses of cross-sectional studies using screening instruments showed a pooled prevalence of 39% for depressive symptoms (95% CI: 30–50% I2: 97%) and 36% for anxiety symptoms (95% CI: 28–44% I2: 97%) among people with epilepsy. Frequencies estimated from screening tools were consistently higher than those based on diagnostic evaluations. Geographically, nearly half of the research was conducted in Asia, followed by North America and Europe.

In summary, the analysis of contributing factors showed that a higher seizure burden and poor seizure control are key drivers of psychiatric symptoms. In fact, experiencing very frequent seizures was associated with a fivefold increase in the odds of major depression. Clinical and electroclinical characteristics also emerged as strong correlates of anxiety and broader psychiatric comorbidity.

Treatment-related factors played an important role as well. Polytherapy and the use of certain antiseizure medications, particularly clonazepam and clobazam, were linked to substantially higher risks of psychiatric symptoms. Demographic and cognitive characteristics, including being female, having an earlier age at epilepsy onset, lower educational attainment, and the presence of cognitive deficits, further heightened vulnerability. Finally, psychosocial and contextual influences were also significant. Perceived stigma, sleep disturbances, and stressors unique to the COVID-19 pandemic, such as disruptions in healthcare services and fear of infection, were all consistently associated

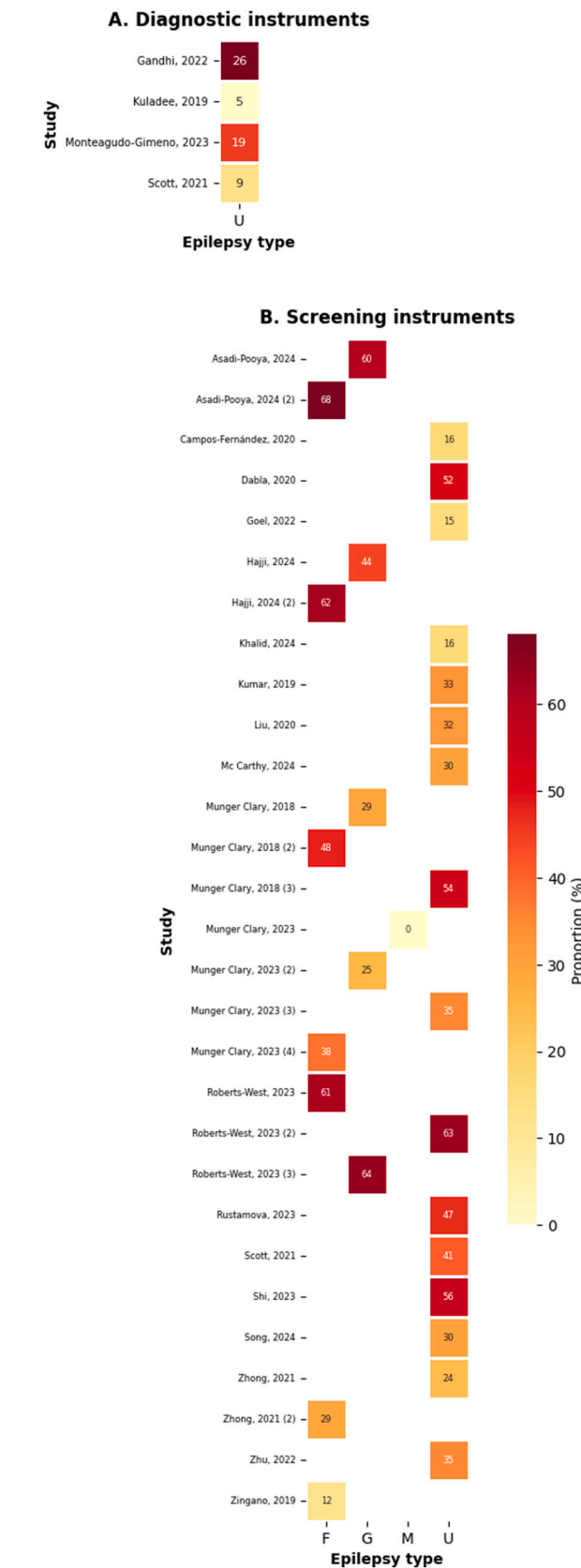


Fig. 4. Anxiety proportions by epilepsy type and ascertainment method (cross-sectional studies).

with increased odds of both depression and anxiety.

At this initial stage, cross-sectional studies revealed a wide range in the reported frequency of depression and anxiety among people with epilepsy. Although most populations across studies were classified under unspecified epilepsy, the few studies that provided stratified data did not show a consistent pattern of predominance across subtypes. This reflects the way information is currently reported in the literature and highlights the need for more systematic subtype-level reporting in future research (Hock et al., 2025).

From the literature, we could say that comparing depression and anxiety across epilepsy subtypes remains challenging because studies classify epilepsy in very different ways. For example, while several reports describe higher rates of depression in temporal lobe epilepsy (Błaszczyk and Czuczwar, 2016; Bosak et al., 2015; Guo et al., 2023), others have found that generalized seizures are linked to greater severity of depressive symptoms (Vacca et al., 2022). Yet another study suggests that people with secondarily generalized seizures may be more prone to depression than those with complex partial seizures (Chandrasekharan et al., 2017). These discrepancies are often attributed to differences, such as the use of diverse psychiatric assessment tools and cut-offs, variations in seizure classification, and differences in antiseizure medication profiles. Similar limitations have been noted by previous scoping review study including population at a first seizure or newly diagnosed epilepsy (Cronin et al., 2023).

Moving beyond the cross-sectional evidence found in our study, analytic studies included also contribute with additional elements to understanding the relationship between PWE and of depression and anxiety. In the broader context, the prevalence of depression and anxiety in adults with epilepsy is substantially higher than in the general population. According to the Health Organization (2025), global prevalence estimates are approximately 5% for depression and 4% for anxiety (World Health Organization [WHO], 2025), whereas our findings show that adults with epilepsy experience frequencies of these symptoms of almost 39% (95% CI: 30%–47% $I^2 = 97.54$) for depressive symptoms and 36% was (95% CI: 28%–44% $I^2 = 96.64\%$) for anxiety symptoms. This pattern aligns with previous reports by Jalal et al. (2023), Fiest et al. (2013), and Scott et al. (2017) (Fiest et al., 2013; Jalal et al., 2023; Scott et al., 2017).

Building on these frequency differences, the studies we reviewed also identified a variety of factors associated with depression and anxiety in people with epilepsy. For clarity, we grouped these into four broad categories: (Fisher et al., 2014) clinical and seizure-related factors, (Sánchez Fernández et al., 2019) treatment-related variables, (Steinmetz et al., 2024) individual characteristics, and (Hajji et al., 2024) psychosocial and contextual influences. Among these, the first category was the one that included most frequently reported factors.

In our review, regarding clinical and seizure-related factors, longer epilepsy duration was reported as a frequent factor associated with depression and anxiety. However, this association must be interpreted cautiously, as the duration of epilepsy varied widely across the included samples, and no consistent clustering around specific stages of the disease was observed. Some previous studies indicate that there are two points in the course of the disease that appear to be associated with a higher incidence of depression: the onset of the disease or its early stages (Al Qahtani et al., 2016; Babbain et al., 2020; De Silva et al., 2021) and long-standing epilepsy (Eyjólfssdóttir et al., 2023; Owolabi et al., 2016). Literature about the duration of epilepsy and anxiety in this regard is also contradictory. Some studies confirm that the duration of epilepsy is a predictor of anxiety (Gavrilovic et al., 2024; Soraya et al., 2022), while others find no relationship between these variables (Balibey et al., 2015; Mensah et al., 2007).

Alongside clinical and seizure-related factors, another commonly reported factor was uncontrolled seizures. Evidence reports that this situation can lead to chronic stress, increased disability, and reduced quality of life, all of which contribute to emotional distress (Niu et al., 2024). Seizure-related variables were the factors most consistently

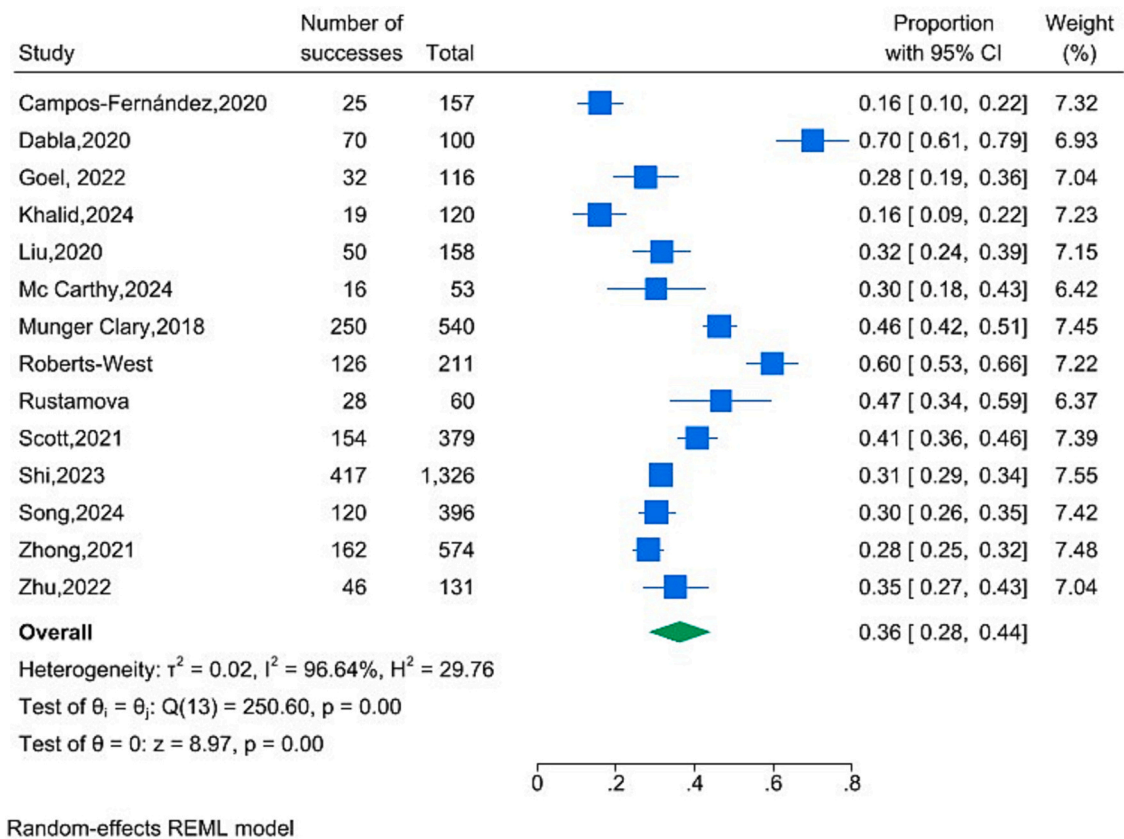


Fig. 5. Forest plot of the meta-analysis of anxiety symptom proportion in individuals with epilepsy.

reported across studies.

Treatment-related factors showed a similar pattern. We found polytherapy was associated with higher levels of anxiety and depression in both men and women. Polytherapy could reflect uncontrolled disease as well as the neuropsychiatric effects of antiseizure medications, particularly when multiple drugs are used simultaneously (Dabla et al., 2020; Choi et al., 2025; Sadek et al., 2025; Berhanu et al., 2025; Han et al., 2025). Within this same category, also patients undergoing epilepsy surgery showed elevated rates of depression and anxiety. Treatment characteristics were consistent among the factors associated with these symptoms.

Regarding individual characteristics, we found that the most of participants in our study were women, although there are no specific results regarding gender and its relationship with anxiety and depression. Previous studies indicate that women with epilepsy are at greater risk of depression or more severe depression than men (Guo et al., 2023; Dabilgou et al., 2019; Hajder et al., 2025; Massi et al., 2025), as well as feeling more anxious (Shi et al., 2023; Niu et al., 2024), while other studies found no differences in relation to depression (Liu et al., 2020; Kimiskidis et al., 2007) or anxiety (Brandt et al., 2010; Lee et al., 2021).

Our study focused on adults with epilepsy, and we did not find evidence suggesting consistent variation in symptoms or diagnoses across different stages of life. However, Cronin et al. (2023) report that anxiety levels can shift throughout adulthood (Cronin et al., 2023), and recent work by Sadek et al. (2025) and Choi et al. (2025) indicates that depressive symptoms may change in intensity and tend to increase in older adults (Choi et al., 2025; Sadek et al., 2025). We found that depression is more frequent with lower educational levels among PWE, a finding that coincides with several studies (Berhanu et al., 2025; Han et al., 2025; Qin et al., 2022; Yang et al., 2020), according to which low educational levels, unemployment, or not being employed are related to up to 3.4 times the risk of depression. Individual characteristics

contributed to the variability observed in the reported findings.

Finally, regarding psychosocial and contextual factors, stigma remains a persistent psychosocial challenge for people with epilepsy and is closely linked to psychiatric comorbidities. In two cross-sectional studies, 31% to 67% of participants reported feeling stigmatized, and this perception showed strong associations with depression, anxiety, and poor medication adherence (Katano and Yamashita, 2022). Higher stigma scores correlated with more severe depressive and anxiety symptoms ($p < 0.01$), and moderate-to-high stigma independently increased the odds of depression (AOR = 3.36) and non-adherence to treatment (AOR = 2.59). Beyond these findings, stigma may also disrupt emotional regulation, delay diagnosis, and contribute to the under-treatment of psychiatric conditions, especially in people with drug-resistant epilepsy (Mula, 2017). Psychosocial factors were likewise frequently noted in association with depressive and anxiety symptoms.

5. Conclusion

Despite marked heterogeneity across studies, our findings show that more than one-third of people with epilepsy experience depressive or anxiety symptoms. Regarding related factors, these conditions remain markedly more frequent in adults with epilepsy than in the general population, driven largely by seizure burden, treatment-related variables, individual vulnerabilities, and psychosocial stressors such as stigma. The wide methodological variation limits the generalizability of pooled frequencies and underscores the need for more standardized assessment and reporting. Overall, these findings emphasize the importance of evaluating people with epilepsy through a comprehensive, holistic approach focused on early psychiatric screening and integrated, multidisciplinary care to improve mental health and quality of life.

CRediT authorship contribution statement

Juan Camilo Rojas-López: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Suhamny Navas-Peñaranda:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Mateo Avilés-Gómez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ximena Palacios-Espinosa:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Leonardo Palacios-Sánchez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Ana María Barragán:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation.

Ethical approval

This study is a systematic review of published literature and did not involve human participants or animals. Therefore, ethical approval was not required.

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Declaration of competing interest

The authors declare no competing financial or personal interests that could have influenced the work reported in this manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jad.2026.121741>.

References

- Abe, C., Denney, D., Doyle, A., Cullum, M., Adams, J., Perven, G., et al., 2020. Comparison of psychiatric comorbidities and impact on quality of life in patients with epilepsy or psychogenic nonepileptic spells. *Epilepsy Behav.* 102, 106649. <https://doi.org/10.1016/j.yebeh.2019.106649>. Jan.
- Adjei, P., Nkromah, K., Akpalu, A., Ohene, S., Puplampu, P., T. Aboagye, E., et al., 2021. Psychological comorbidities in epilepsy: a cross-sectional survey among Ghanaian epilepsy patients. *Ghana Med. J.* 55 (2), 128–134. <https://doi.org/10.4314/gmj.v55i2.5>. Jun 1.
- Ajinkya, S., Fox, J., Lekoubou, A., 2020. Trends in prevalence and treatment of depressive symptoms in adult patients with epilepsy in the United States. *Epilepsy Behav.* 105, 106973. <https://doi.org/10.1016/j.yebeh.2020.106973>. Apr.
- Al Qahtani, M., Al Qahtani, D., Al Malwi, F., Assiri, M., Velmurugan, M., Bhatia, H., et al., 2016. Depression in epileptic patients is influenced by the duration of epilepsy: a cross sectional study in patients with epilepsy (P4.209). *Neurology* 86 (16, supplement). https://doi.org/10.1212/WNL.86.16_supplement.P4.209. Apr 5.
- Babtain, F.A., Alqahtani, M., AlQahtani, D.H., Al Malwi, A.S., Bahtia, H., et al., 2020. Depression in patients with epilepsy is influenced by the duration of epilepsy. *Am. J. Med. Med. Sci.* 10 (6), 422–429. <https://doi.org/10.5923/j.ajmms.20201006.15>.
- Balibey, H., Yasar, H., Tekeli, H., Bayar, N., 2015. Frequency of anxiety and depression in epileptic patients. *Klin Psikofarmakol Bül-Bull. Clin. Psychopharmacol.* 25 (2), 136–140. <https://doi.org/10.5455/bcp.20130429122553>. Jun.
- Berg, A.T., Altalib, H.H., Devinsky, O., 2017. Psychiatric and behavioral comorbidities in epilepsy: a critical reappraisal. *Epilepsia* 58 (7), 1123–1130. <https://doi.org/10.1111/epi.13766>. Jul.
- Berhanu, T., Yizgaw, N., Tinsae, S., Mirkena, Y., Belete, A., 2025. Depression and associated factors among people with epilepsy at Wolaita Sodo University hospital, South Ethiopia. *Discov. Ment. Health* 5 (1), 33. <https://doi.org/10.1007/s44192-025-00161-1>. Mar 10.
- Błaszczak, B., Czuczwar, S.J., 2016. Epilepsy coexisting with depression. *Pharmacol. Rep.* 68 (5), 1084–1092. <https://doi.org/10.1016/j.pharep.2016.06.011>. Oct.
- Bosak, M., Dudek, D., Siwek, M., Szczudlik, A., 2015. Subtypes of interictal depressive disorders according to ICD-10 in patients with epilepsy. *Neurol. Neurochir. Pol.* 49 (2), 90–94. <https://doi.org/10.1016/j.pjnns.2015.01.008>. Mar.
- Brandt, C., Schoendienst, M., Trentowska, M., May, T.W., Pohlmann-Eden, B., Tuschen-Caffier, B., et al., 2010. Prevalence of anxiety disorders in patients with refractory focal epilepsy—a prospective clinic based survey. *Epilepsy Behav.* 17 (2), 259–263. <https://doi.org/10.1016/j.yebeh.2009.12.009>. Feb.
- Campos-Fernández, D., Fonseca, E., Olivé-Gadea, M., Quintana, M., Abaira, L., Seijo-Raposo, I., et al., 2020. The mediating role of epileptic seizures, irritability, and depression on quality of life in people with epilepsy. *Epilepsy Behav.* 113, 107511. <https://doi.org/10.1016/j.yebeh.2020.107511>. Dec.
- Chandrasekharan, S.C., Menon, V., Wadwekar, V., Nair, P.P., 2017. High frequency of depressive symptoms among adults with epilepsy: results from a hospital-based study. *J. Neurosci. Rural Pract.* 08 (S 01), S013–S019. <https://doi.org/10.4103/jnrp.jnrp.21.17>. Aug.
- Choi, H., Stepan, B., Clifton, R., Longstreth, W.T., Elkind, M.S.V., Gutierrez, J., et al., 2025. Association of epilepsy with trajectory of depressive symptoms in late life: the cardiovascular health study. *Epilepsia* 66 (12), 4834–4843. <https://doi.org/10.1111/epi.18579>. Dec.
- Cronin, W., Kwan, P., Foster, E., 2023. Anxiety and depressive symptoms in adults with new-onset seizures: a scoping review. *Epilepsia Open.* 8 (3), 758–772. <https://doi.org/10.1002/epi4.12766>. Sep.
- Dabilgou, A.A., Nanema, D., Dravé, A., Sawadogo, S.P., Kyelem, J.M.A., Napon, C., et al., 2019. Symptoms of depression and associated risk factors in patients with epilepsy in Burkina Faso. *Open J. Depress.* 08 (01), 29–40. <https://doi.org/10.4236/ojd.2019.81004>.
- Dabla, S., Juneja, H., Sing, P., Bala, K., 2020. Depression, anxiety and suicidal ideation/behaviour among persons with epilepsy: common but underestimated comorbidities in Haryana, North India. *Neurol. Asia* 25 (1), 7–12.
- De Silva, S., Isuru, A., Rodrigo, A., Kuruppuarachchi, L., 2021. Prevalence and correlates of depression in patients with epilepsy in Sri Lanka. *Ceylon Med. J.* 66 (3), 138. <https://doi.org/10.4038/cmj.v66i3.9492>. Dec 27.
- Eadie, M.J., 2011. William Gowers' interpretation of epileptogenic mechanisms: 1880–1906: Gowers and epileptogenic mechanisms. *Epilepsia* 52 (6), 1045–1051. <https://doi.org/10.1111/j.1528-1167.2011.02983.x>. Jun.
- Eyjólfssdóttir, S.G., Trinka, E., Höller, Y., 2023. Shorter duration of slow wave sleep is related to symptoms of depression in patients with epilepsy. *Epilepsy Behav.* 149, 109515. <https://doi.org/10.1016/j.yebeh.2023.109515>. Dec.
- Faught, E., Besson, H., D'Souza, W., Klein, P., Reuber, M., Rosenow, F., et al., 2024. Brivaracetam effectiveness and tolerability in older and younger adults with epilepsy: EXPERIENCE, a pooled analysis of international data from retrospective studies. *Epilepsy Behav.* 158, 109922. <https://doi.org/10.1016/j.yebeh.2024.109922>. Sep.
- Fiest, K.M., Dykeman, J., Patten, S.B., Wiebe, S., Kaplan, G.G., Maxwell, C.J., et al., 2013. Depression in epilepsy: a systematic review and meta-analysis. *Neurology* 80 (6), 590–599. <https://doi.org/10.1212/WNL.0b013e31827b1ae0>. Feb 5.
- Fisher, R.S., Acevedo, C., Arzimanoglou, A., Bogacz, A., Cross, J.H., Elger, C.E., et al., 2014. ILAE official report: a practical clinical definition of epilepsy. *Epilepsia* 55 (4), 475–482. <https://doi.org/10.1111/epi.12550>. Apr.
- Gandhi, S., Menon, P., Jagtap, B.L., Chaudhury, S., Saldanha, D., 2022. Prevalence and correlates of psychiatric comorbidity in patients with epilepsy: a Cross-sectional hospital-based study. *Med. J. Dr. Patil. Vidyapeeth.* 15 (6), 881–890. <https://doi.org/10.4103/mjdrdypu.mjdrdypu.654.20>. Nov.
- Gandy, M., Wu, W., Woldhuis, T., Bennett, S.D., Baslet, G., Araujo-Filho, G., et al., 2025. Integrated care for mental health in epilepsy: a systematic review and meta-synthesis by the international league against epilepsy integrated mental health care pathways task force. *Epilepsia* 66 (4), 1024–1040. <https://doi.org/10.1111/epi.18252>. Apr.
- Gavrilovic, A., Gavrilovic, J., Ilic Zivojinovic, J., Jeličić, L., Radovanovic, S., Vesic, K., 2024. Influence of epilepsy characteristics on the anxiety occurrence. *Brain Sci.* 14 (9), 858. <https://doi.org/10.3390/brainsci14090858>. Aug 26.
- Goel, P., Singh, G., Bansal, V., Sharma, S., Kumar, P., Chaudhry, R., et al., 2022. Psychiatric comorbidities among people with epilepsy: a population-based assessment in disadvantaged communities. *Epilepsy Behav.* 137, 108965. <https://doi.org/10.1016/j.yebeh.2022.108965>. Dec.
- Guo, W., Li, Y., xuan, Zhang, Y., Lv, X. rui, Wang, S. xue, Zhang, S. yuan, et al., 2023. Risk analysis of depression among adult patients with epilepsy of different sex: a retrospective single-center study from China. *Front. Psych.* 14, 1283983. <https://doi.org/10.3389/fpsy.2023.1283983>. Dec 4.
- Hajder, D., Sekulić, S., Ignjatović, V.B., Popović, S., Popović, N., Nikolašević, Ž., et al., 2025. Clinical and sociodemographic predictors of depressive symptoms in epilepsy patients in a single tertiary epilepsy center. *Acta Neurol. Belg.* 125 (1), 89–98. <https://doi.org/10.1007/s13760-024-02632-8>. Feb.

- Hajji, E.B., Traore, B., Hassoune, S., Khoubila, A., Bellakhdar, S., Rafai, M.A., et al., 2024. Prevalence and associated factors of psychiatric comorbidities in epilepsy in the Casablanca-Settat region of Morocco: a cross-sectional study. *Epilepsia Open*. 9 (4), 1321–1332. <https://doi.org/10.1002/epi4.12958>. Aug.
- Han, Y., Hao, G., Chu, G., Yang, D., Sun, S., Gong, S., et al., 2025. Global, regional and national burdens of epilepsy in the adolescents and young adults from 1990 to 2021 and its predictions. *BMC Neurol*. 25 (1), 402. <https://doi.org/10.1186/s12883-025-04331-0>. Sep 30.
- Ho, P., Leung, W.C., Leung, I.Y., Chang, R.S., 2020. Factors associated with depression in people with epilepsy: a retrospective case-control analysis. *Hong Kong Med. J.* <https://doi.org/10.12809/hkmj198310>. Jul 2.
- Hock, S., Tako, L.M., Knake, S., Von Podewils, F., Langenbruch, L., Kovac, S., et al., 2025. Occurrence and treatment of depression symptoms and anxiety in patients with epilepsy: findings from a German multicenter cross-sectional study. *Epilepsy Behav.* 173, 110570. <https://doi.org/10.1016/j.yebeh.2025.110570>. Dec.
- Jalal, O., Ahmad, U., Siddique Kakar, M., Muhammad, A., Fatima, M., Hammad, S., 2023. Frequency of anxiety and depression in patients of epilepsy. *Pak. J. Med. Health Sci.* 17 (10). <https://doi.org/10.53350/pjmhs20231710150>. Oct 28.
- Jansen, C., Francome, L., Vignal, J.P., Jacquot, C., Schwan, R., Tyvaert, L., et al., 2019. Interictal psychiatric comorbidities of drug-resistant focal epilepsy: prevalence and influence of the localization of the epilepsy. *Epilepsy Behav.* 94, 288–296. <https://doi.org/10.1016/j.yebeh.2018.06.046>. May.
- Katano, A., Yamashita, H., 2022. The impact of palliative radiation therapy on patients with advanced gastric Cancer: results of a retrospective cohort study. *Cureus*. <https://doi.org/10.7759/cureus.32971>. Dec 26.
- Katyal, J., Rashid, H., Tripathi, M., Sood, M., 2021. Prevalence of depression and suicidal ideation in persons with epilepsy during the COVID-19 pandemic: a longitudinal study from India. *Epilepsy Behav.* 124, 108342. <https://doi.org/10.1016/j.yebeh.2021.108342>. Nov.
- Khalid, B., Waqar, Z., Khan, S., Ali, I., Afzal, N., Irfan, A., et al., 2024. Psychiatric implications of anti-seizure medications in epileptic population. *BMC Neurol*. 24 (1), 166. <https://doi.org/10.1186/s12883-024-03670-8>. May 21.
- Kimikidis, V.K., Triantafyllou, N.I., Kararizou, E., Gatzonis, S.S., Fountoulakis, K.N., Siatouni, A., et al., 2007. Depression and anxiety in epilepsy: the association with demographic and seizure-related variables. *Ann. General Psychiatry* 6 (1), 28. <https://doi.org/10.1186/1744-859X-6-28>. Dec.
- Kuladee, S., Prachason, T., Srisopit, P., Trakulchang, D., Boongird, A., Wisajan, P., et al., 2019. Prevalence of psychiatric disorders in Thai patients with epilepsy. *Epilepsy Behav.* 90, 20–24. <https://doi.org/10.1016/j.yebeh.2018.11.004>. Jan.
- Lee, S.A., Cho, Y.J., Ryu, H.U., Kim, K.T., Seo, J.G., Kang, K.W., et al., 2021. Sex differences in seizure effects on social anxiety in persons with epilepsy. *Epilepsy Behav.* 124, 108318. <https://doi.org/10.1016/j.yebeh.2021.108318>. Nov.
- Li, P., Lin, J., Wu, C., Huang, S., Zhu, S., 2021. The impact of social factors, especially psychological worries on anxiety and depression in patients with epilepsy. *Epilepsy Behav.* 125, 108376. <https://doi.org/10.1016/j.yebeh.2021.108376>. Dec.
- Liu, Z., Yin, R., Fan, Z., Fan, H., Wu, H., Shen, B., et al., 2020. Gender differences in associated and predictive factors of anxiety and depression in people with epilepsy. *Front. Psych.* 11, 670. <https://doi.org/10.3389/fpsy.2020.00670>. Jul 10.
- Livingston, S., 1972. *Comprehensive Management of Epilepsy in Infancy, Childhood and Adolescence*. Charles C Thomas.
- Manole, A., Sirbu, C., Mititelu, M., Vasiliu, O., Lorusso, L., Sirbu, O., et al., 2023. State of the art and challenges in epilepsy—a narrative review. *J. Pers. Med.* 13 (4), 623. <https://doi.org/10.3390/jpm13040623>. Apr 1.
- Massi, D.G., Feudjio, R., Eyoum, C., Paternoster, L., Magnerou, A.M., Ferreira, N.T., et al., 2025. Prevalence of depression in people with epilepsy: a hospital-based study in Cameroon. *Epilepsy Behav.* 165, 110326. <https://doi.org/10.1016/j.yebeh.2025.110326>. Apr.
- Mensah, S.A., Beavis, J.M., Thapar, A.K., Kerr, M.P., 2007. A community study of the presence of anxiety disorder in people with epilepsy. *Epilepsy Behav.* 11 (1), 118–124. <https://doi.org/10.1016/j.yebeh.2007.04.012>. Aug.
- Monteagudo-Gimeno, E., Sánchez-González, R., Raduà-Castaño, J., Fortea-González, L., Boget-Llucía, T., Carreño-Martínez, M., et al., 2023. Clinical relevance of interictal dysphoric disorder and its impact on quality of life in drug-resistant epilepsy. *Epilepsy Behav.* 144, 109253. <https://doi.org/10.1016/j.yebeh.2023.109253>. Jul.
- Moola, S., Munn, Z., Tufanaru, C., Aromataris, E., Sears, K., Sfetcu, R., et al., 2017. Checklist for case control studies. In: Aromataris, E., Munn, Z. (Eds.), *JB1 Manual for Evidence Synthesis* [Internet]. Johanna Briggs Institute JBI, pp. 1–6. Available from: <https://synthesismanual.jbi.global>.
- Moola, S., Munn, Z., Tufanaru, C., Aromataris, E., Sears, K., Sfetcu, R., et al., 2020a. Checklist For Analytical Cross Sectional Studies Critical Appraisal tools for use in JBI Systematic Reviews. In: Aromataris, E., Munn, Z. (Eds.), *JB1 Manual for Evidence Synthesis* [Internet]. Johanna Briggs Institute JBI. Available from: <https://synthesismanual.jbi.global>.
- Moola, S., Munn, Z., Tufanaru, C., Aromataris, E., Sears, K., Sfetcu, R., et al., 2020b. Checklist for cohort studies. In: Aromataris, E., Munn, Z. (Eds.), *JB1 Manual for Evidence Synthesis* [Internet]. Johanna Briggs Institute JBI, pp. 1–6. Available from: <https://synthesismanual.jbi.global>.
- Mula, M., 2017. Depression in epilepsy. *Curr. Opin. Neurol.* 30 (2), 180–186. <https://doi.org/10.1097/WCO.0000000000000431>. Apr.
- Munger Clary, H.M., Snively, B.M., Hamberger, M.J., 2018. Anxiety is common and independently associated with clinical features of epilepsy. *Epilepsy Behav.* 85, 64–71. <https://doi.org/10.1016/j.yebeh.2018.05.024>. Aug.
- Munger Clary, H.M., Giambardi, L., Floyd, W.N., Hamberger, M.J., 2023. Afraid to go out: poor quality of life with phobic anxiety in a large cross-sectional adult epilepsy center sample. *Epilepsy Res.* 190, 107092. <https://doi.org/10.1016/j.epilepsyres.2023.107092>. Feb.
- Niu, C., Li, P., Du, X., Zhao, M., Wang, H., Yang, D., et al., 2024. Risk factors for anxiety in patients with epilepsy: a meta-analysis. *Epilepsy Behav.* 153, 109665. <https://doi.org/10.1016/j.yebeh.2024.109665>. Apr.
- Ogundare, T., Adebawale, T.O., Borba, C.P.C., Henderson, D.C., 2020. Correlates of depression and quality of life among patients with epilepsy in Nigeria. *Epilepsy Res.* 164, 106344. <https://doi.org/10.1016/j.epilepsyres.2020.106344>. Aug.
- Owolabi, D., Owolabi, L., Udofia, O., Sale, S., 2016. Depression in clinically diagnosed patients with epilepsy in northwestern Nigeria (P4.207). *Neurology* 86 (16, supplement). https://doi.org/10.1212/WNL.86.16_supplement.P4.207. P4.207. Apr 5.
- Palacios Espinosa, X., Arévalo Quinchaneque, J.J., Palacios, Sánchez L., 2024. Reflexiones sobre la estigmatización de la epilepsia a través de la historia. *Salud Uninorte* 40 (03), 1093–1112. <https://doi.org/10.14482/sun.40.03.654.456>. Dec 17.
- Qin, S. kun, Yang, Z. xian, Guan, Z. wei, Zhang, J. hu, Ping, X., Lu, Y., et al., 2022. Exploring the association between epilepsy and depression: a systematic review and meta-analysis. *Farrukh MJ, editor. PLoS One* 17 (12), e0278907. <https://doi.org/10.1371/journal.pone.0278907>. Dec 15.
- Quiroga-Padilla, P.J., Briceño, C., Mayor, L.C., 2022. Factors associated with initiation of the modified Atkins diet in adults with drug-resistant epilepsy. *Epilepsy Behav.* 129, 108620. <https://doi.org/10.1016/j.yebeh.2022.108620>. Apr.
- Sadek, M., Chemali, Z., Sarkis, R.A., 2025. Depression in older adults with epilepsy: pathophysiology and management considerations. *Epilepsy Behav.* 172, 110692. <https://doi.org/10.1016/j.yebeh.2025.110692>. Nov.
- Saeedi, N., Hatamian, H., Erfanian, M., Shafiei, F., Homam, S.M., 2024. Mental health and seizure patterns in epileptic patients during COVID-19 pandemic: a case study in Northeast Iran. *Casp. J. Neurol. Sci.* 10 (3), 225–234. <https://doi.org/10.32598/CJNS.10.38.389.4>. Jul 1.
- Sánchez Fernández, I., Goodkin, H.P., Scott, R.C., 2019. Pathophysiology of convulsive status epilepticus. *Seizure* 68, 16–21. <https://doi.org/10.1016/j.seizure.2018.08.002>. May.
- Scott, A.J., Sharpe, L., Hunt, C., Gandy, M., 2017. Anxiety and depressive disorders in people with epilepsy: a meta-analysis. *Epilepsia* 58 (6), 973–982. <https://doi.org/10.1111/epi.13769>. Jun.
- Scott, A.J., Sharpe, L., Thayer, Z., Miller, L.A., Nikpour, A., Parratt, K., et al., 2021. How frequently is anxiety and depression identified and treated in hospital and community samples of adults with epilepsy? *Epilepsy Behav.* 115, 107703. <https://doi.org/10.1016/j.yebeh.2020.107703>. Feb.
- Sehlo, M.G., Mohamed, W.S., Youssef, U.M., Lotfi, S.E., El-deen, G.M.S., 2022. Prevalence and determinants of depression in patients with epilepsy during the COVID-19 pandemic. *Middle East Curr. Psychiatry* 29 (1), 22. <https://doi.org/10.1186/s43045-022-00190-4>. Dec.
- Shi, W., Sun, H., Peng, W., Chen, Z., Wang, Q., Lin, W., et al., 2023. Prevalence and risk factors of anxiety and depression in adult patients with epilepsy: a multicenter survey-based study. *Ther. Adv. Neurol. Disord.* 16. <https://doi.org/10.1177/17562864231187194>. Jan.
- Soraya, H., Amin, M.M., Husada, M.S., Effendy, E., 2022. Factors associated with anxiety scores in people with epilepsy in neurological installations of haji Adam Medan center general hospital. *Open Access Maced. J. Med. Sci.* 10 (T7), 108–112. <https://doi.org/10.3889/oamjms.2022.9255>. Apr 16.
- Steinmetz, J.D., Seehar, K.M., Schiess, N., Nichols, E., Cao, B., Servili, C., et al., 2024. Global, regional, and national burden of disorders affecting the nervous system, 1990–2021: a systematic analysis for the global burden of disease study 2021. *Lancet Neurol.* 23 (4), 344–381. [https://doi.org/10.1016/S1474-4422\(24\)00038-3](https://doi.org/10.1016/S1474-4422(24)00038-3). Apr.
- Sylla, M., Vogel, A.C., Bah, A.K., Tassiou, N.R., Barry, S.D., Djibo, B.A., et al., 2020. Prevalence, severity, and associations of depression in people with epilepsy in Guinea: a single-center study. *Epilepsy Behav.* 113, 107475. <https://doi.org/10.1016/j.yebeh.2020.107475>. Dec.
- Talibov, T., İnci, M., İsmayilov, R., Elmas, S., Büyüktopçu, E., Kepenek, A.O., et al., 2024. The relationship of psychiatric comorbidities and symptoms, quality of life, and stigmatization in patients with epilepsy. *Epilepsy Behav.* 156, 109838. <https://doi.org/10.1016/j.yebeh.2024.109838>. Jul.
- Tandiawan, F., Gunadharma, S., Gamayani, U., Ong, A., Rizal Ganiem, A., Sobaryati, et al., 2023. Depression as comorbidity in Sundaese epilepsy patients. *Neurol. Asia* 28 (4), 935–941. <https://doi.org/10.54029/2023xmy>. Dec.
- Vacca, M., Fernandes, M., Spanetta, M., Placidi, F., Izzi, F., Lombardo, C., et al., 2022. Depressive symptoms in patients with epilepsy and clinically associated features in a single tertiary center. *Neurol. Sci.* 43 (3), 1965–1974. <https://doi.org/10.1007/s10072-021-05589-1>. Mar.
- World Health Organization [WHO], 2025. Depressive Disorder (Depression) [Internet]. World Health Organization. Available from: <https://www.who.int/news-room/fact-sheets/detail/depression>.
- Yang, Y., Yang, M., Shi, Q., Wang, T., Jiang, M., 2020. Risk factors for depression in patients with epilepsy: a meta-analysis. *Epilepsy Behav.* 106, 107030. <https://doi.org/10.1016/j.yebeh.2020.107030>. May.
- Zhong, R., Chen, Q., Li, M., Li, N., Zhang, X., Lin, W., 2021. Sex differences in anxiety in patients with epilepsy: status and risk factors analysis. *Epilepsy Behav.* 116, 107801. <https://doi.org/10.1016/j.yebeh.2021.107801>. Mar.